

Aggressive feature reduction degrades the selective reliability and clinical net benefit of clinical risk models

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ABSTRACT

Feature reduction is widely adopted in clinical risk modelling, usually justified by negligible loss of discrimination (AUROC). Deployment relies on more: clinicians act on calibrated probabilities, triage must order its own errors, and equitable care requires reliability across subgroups. We introduce a pre-registered, leakage-safe protocol certifying reduced models on these properties, applied to twelve public clinical datasets ($n = 155\text{--}101,763$), three rankers, three learners, and budgets from 100 % to 25 % plus matched exact- k budgets ($k \in \{1, 2, 4, 8\}$), with paired, multiplicity-corrected, bootstrapped comparisons. With a quarter retained, degradation was material in most of 108 dataset–ranker–learner cells: selective reliability worse in 94, net benefit in 80, conformal set size in 79, discrimination in 91; calibration error and subgroup gaps were mixed. Two-thirds of the selective-reliability loss reflected the probability scale; a third persisted under a feature-space confidence. The harm tracked the *absolute* retained count: at $k = 1$ every dataset was materially harmed, and reduction flipped 21 % of biopsy decisions on mammographic data at threshold 0.10. Recalibration across five methods of increasing flexibility repaired calibration error and recovered no selective reliability. Discrimination is an insufficient safety certificate: calibration, selective reliability, clinical utility, and subgroup equity should be audited per dataset, learner, and retained count.

1 Introduction

Predictive models built on tabular clinical data—heart-disease risk, diabetes onset, liver-disease screening, post-heart-failure mortality—are typically reported with a single headline of discrimination, most often the area under the receiver-operating characteristic curve (AUROC). When such models are deployed as decision support, however, discrimination is not the primary property a clinician relies on. A clinician reads a *probability*—“this patient has an 18 % risk”—and the value of that number depends on whether it is *calibrated*: whether, among patients assigned an 18 % risk, close to 18 % actually experience the event. A model can rank patients well (high AUROC) and yet be badly calibrated, systematically over- or under-stating absolute risk, which is precisely the failure mode that misleads treatment thresholds.

Feature reduction is an almost universal preprocessing step. Reducing the feature budget lowers data-collection cost and simplifies deployment, and is typically defended by showing that AUROC changes little when the model is restricted to the top-ranked features. This accuracy-centred justification, however, does not examine two properties of direct clinical importance: calibration and *subgroup reliability*—whether calibration and discrimination hold equally across protected groups such as sex or age. These properties are seldom re-examined after features assessed to be redundant are dropped. If aggressive reduction degrades calibration or widens a subgroup gap while leaving AUROC almost intact, a model selected on the conventional evidence is less trustworthy and less equitable than its developers might conclude from that evidence.

This paper addresses that risk with a methodological contribution: not a new feature-selection method, but a reusable way to *certify* a reduced model. In the prevailing practice, feature reduction is certified solely on discrimination (AUROC); we replace this with a pre-registered, leakage-safe protocol that evaluates a reduced model on the properties a deployed risk score is actually trusted for—calibration, selective (risk–coverage) reliability, and between-subgroup equity. The protocol quantifies, as the fraction of retained features decreases, the change in calibration (expected calibration error, ECE, and the Brier score), in selective reliability (the area under the risk–coverage curve, AURC, which captures whether the model’s own confidence can be trusted to triage which predictions to act on), in discrimination (AUROC), and in the between-subgroup gaps in AUROC and calibration. We fix the outcomes, the reduction levels, and the decision rule before examining any result, apply the protocol under a strict leakage firewall to twelve public datasets with three base learners and three ranking methods, and report results in both directions: the settings in which reduction is safe and those in which it is not.

Contributions. (i) A reusable, pre-registered, leakage-safe evaluation protocol that certifies a feature-reduced clinical risk model on calibration, selective reliability, and subgroup equity rather than on discrimination alone—the current state-of-the-art justification for feature reduction—with every number regenerable from committed code and recorded seeds. (ii) An instantiation across twelve public clinical datasets, three learners, and three rankers, showing the protocol reveals deployment-relevant degradation that discrimination-only evaluation misses, and that the harm is strongly conditional on dataset and learner—with an exploratory cross-dataset meta-analysis that turns this conditionality into an *a priori* screen for when reduction is safe. (iii) A leakage-safe recalibration control, built into the protocol, that separates *repairable* (calibration-error) from *irreparable* (selective-reliability, Brier) harm, yielding an actionable, both-directions decision rule for practitioners.

2 Related work

Feature selection in clinical machine learning. Filter, embedded, and wrapper feature-selection methods are standard in clinical prediction, and are conventionally evaluated by discrimination or classification accuracy on the selected set of features^{1,2}. We use three widely adopted rankers as representative of this practice—a filter (mutual information), an embedded tree-importance ranker (random-forest Gini importance³), and an embedded sparse linear ranker (L_1 -penalised logistic regression)—and treat the *ranking* as given, assessing the loss due to its aggressive application.

Precedent for reduction on low-dimensional clinical data. Although feature selection is most often associated with high-dimensional problems, aggressive reduction is routinely applied to clinical tabular data of exactly the dimensionality studied here, for reasons of collection cost, parsimony, and bedside usability rather than statistical necessity: classical risk scores distill a handful of predictors from larger candidate sets, and the practice persists in the machine-learning literature on these same datasets. A prominent example is the survival study on the heart-failure cohort we reuse, which argued that two features (serum creatinine and ejection fraction) suffice out of a twelve-feature candidate set¹⁷—a reduction to a sixth of the features, justified by discrimination-style evidence alone, on a dataset our audit places in the harmed, compact regime. The Cleveland and Pima datasets were likewise introduced with compact probability models built on selected subsets of the recorded variables^{14,16}. This is precisely the regime our protocol audits; the high-dimensional regime is examined separately in Section 4.8 and remains outside the paper’s guidance, for reasons the measurements there make concrete rather than assumed.

Calibration of risk models. Calibration—the agreement between predicted probabilities and observed frequencies—is a long-standing requirement for clinical risk scores, quantified by the Brier score⁴ and by binned expected calibration error^{5,6}. It is well established that good discrimination does not imply good calibration⁷, and that prediction models should be judged by a set of measures that span discrimination, calibration, and clinical utility⁸. Our contribution is to treat calibration as a *dependent variable of preprocessing*: we measure how it is altered as features are removed, rather than measuring it only for the final model.

Selective prediction and the risk–coverage trade-off. Selective (reject-option) classification lets a model *abstain*—decline to issue a prediction—on the inputs it is least sure about⁹; its quality is summarised by the risk–coverage curve and its area (AURC), which is low when the model’s confidence ranks its errors well¹⁰. We include AURC because it captures a deployment-relevant property—can the model be trusted to triage its own predictions?—that neither AUROC nor a single calibration number captures.

Algorithmic fairness in health. Subgroup performance gaps—differences in discrimination or calibration across protected groups such as sex or age—are a central concern for equitable clinical AI^{11,12}. Most audits fix the model and compare groups; instead, we ask whether a routine engineering choice (feature reduction) *widens* such gaps, connecting the fairness and preprocessing literatures.

Gap. To our knowledge these threads are usually not combined: feature reduction is justified by accuracy, while calibration, selective reliability, and subgroup gaps are audited—if at all—on the final model only. We close that gap with a single pre-registered protocol that tracks all four attributes as a function of the feature budget.

3 Methods

We formalise the audit as a reusable certification protocol whose architecture is summarised in Fig. 1 and specified in Algorithms 1 and 2. Sections 3.1–3.5 define each component; Section 4 applies the protocol to twelve datasets, three learners, and three rankers. The protocol takes a clinical dataset and a feature-reduction configuration and returns, for every retained-feature budget, a direction-resolved verdict on each reliability axis, computed under a strict leakage firewall so that no information from a held-out patient can enter preprocessing. The protocol is designed to operationalise the calibration, clinical-utility, and fairness reporting items of TRIPOD+AI, the reporting standard for clinical prediction models using machine

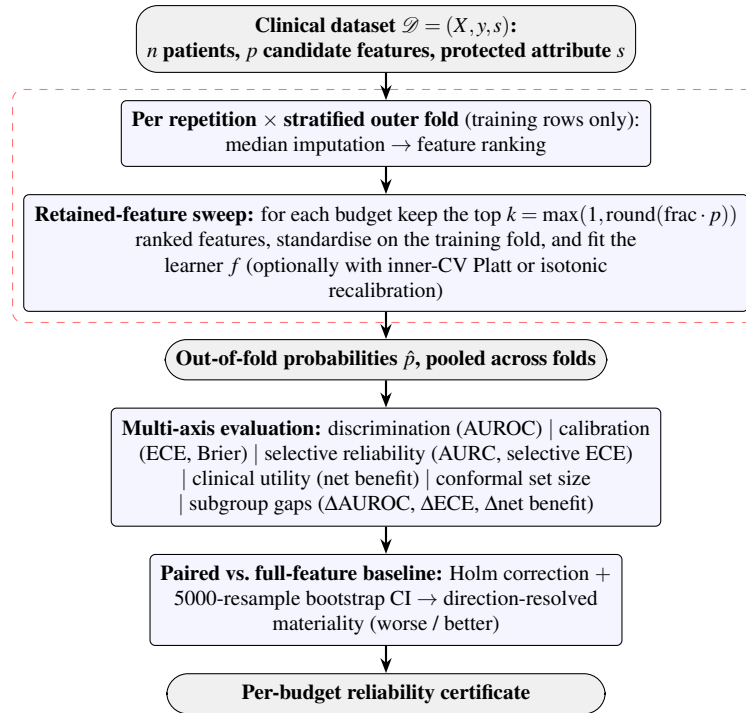


Figure 1. Architecture of the certification protocol. Imputation, feature ranking, standardisation, and model fitting all occur inside the training portion of each cross-validation fold (the dashed *leakage firewall*); only out-of-fold probabilities are scored. The budget sweep covers both fractional budgets and matched exact- k budgets, which share the same folds and rankings. Each reliability axis is compared, paired across repetitions, to the full-feature baseline and resolved into a materially-worse or materially-better verdict per budget.

learning¹³: it reports calibration (ECE, Brier) and clinical utility (decision-curve net benefit) alongside discrimination, examines performance across protected subgroups, and releases code and a pre-registration so that every reported quantity is reproducible. A completed adherence note is provided in the repository.

3.1 Datasets

We use twelve public clinical tabular datasets (Table 1), spanning cardiology (Cleveland¹⁴, Statlog and SPECTF heart), oncology (the Wisconsin diagnostic¹⁵ and original breast-cancer datasets, mammographic mass, and Haberman survival), metabolic and hepatic disease (Pima diabetes¹⁶, the Indian Liver Patient Dataset, hepatitis), post-heart-failure mortality¹⁷, and a large real-world electronic-health-record cohort, the Diabetes 130-US hospitals dataset¹⁸ (early readmission). Each carries a binary outcome and, where available, a protected attribute that is *never* used as a model feature. The datasets span three orders of magnitude in sample size ($n = 155\text{--}101,763$; the readmission cohort is analysed in full), feature counts from $p = 3$ to 44, outcome prevalences from 0.11 to 0.79, and protected attributes covering sex, gender, age, and (in the readmission cohort) race, so the audit is not tied to one clinical context and supports a cross-dataset meta-analysis (Section 4.7). Throughout the paper we call a dataset *compact* when its candidate set is small ($p \leq 12$, so the most aggressive budget retains $k \leq 3$ features) and *wide* when it is large ($p \geq 16$, retaining $k \geq 4$ at the same budget); the meta-analysis figure marks each dataset’s class.

Clinical stakes. Each outcome is acted on at a probability threshold, which is precisely why calibration and selective reliability—not ranking alone—matter clinically. On Cleveland, coronary-disease risk informs statin initiation and further cardiac testing, so a reduced model that over- or under-states absolute risk shifts which patients are treated; sex is the equity axis. Pima diabetes-onset risk drives preventive-intervention and screening decisions, where a deflated risk misses preventable onset; here the calibration–fairness gap is examined across an age stratum. ILPD liver-disease risk informs specialist referral, with gender as the equity axis. Heart-Failure mortality risk informs intensity-of-care and palliative-care discussions—a setting in which a model used to triage must properly order its own errors (AURC). WDBC malignancy probability governs whether a borderline case is flagged for human review; it carries no protected attribute and serves as the wide-feature, reduction-benign anchor. The Diabetes 130-US readmission cohort adds a large electronic-health-record setting in which a 30-day-readmission risk informs discharge and care-transition decisions, with *race* as the equity axis across six groups—the most consequential

Table 1. Datasets. Prevalence is the positive-class rate; p is the number of model features (the protected attribute is excluded from the features). All are publicly available; see Data availability.

Dataset	Outcome	n	p	Prev.	Protected
bcw	malignant	699	9	0.345	none
cleveland	coronary disease	303	12	0.459	sex
diabetes130	30-day readmission	101,763	16	0.112	race
haberman	5-year death	306	3	0.265	age group
heartfailure	death	299	11	0.321	sex
hepatitis	death	155	18	0.206	sex
ilpd	liver disease	583	9	0.714	gender
mammographic	malignant	961	5	0.463	age group
pima	diabetes onset	768	8	0.349	age group
spectf	abnormal SPECT	267	44	0.794	none
statlogheart	coronary disease	270	12	0.444	sex
wdbc	malignant	569	30	0.373	none

fairness context in the study. The remaining datasets (Statlog and SPECTF heart, mammographic mass, Haberman survival, hepatitis, original breast cancer) apply the same threshold-decision logic to further diagnostic and prognostic tasks. Across these contexts, a miscalibrated reduced model, or one whose confidence no longer orders its errors, would misallocate the very decisions the score is intended to support.

3.2 Leakage-controlled protocol

For each combination of dataset, ranking method, base learner, and repetition, we run stratified five-fold cross-validation; throughout this paper k denotes only the number of *retained features*. Within *every* training fold, and using training data only, we (i) impute missing values by the training-fold median, (ii) rank all features by the chosen method, and (iii) for each retained fraction keep the top- k features ($k = \max(1, \text{round}(\text{frac} \times p))$, with round-half-to-even), standardise them on the training fold, and fit the base learner. Two properties of this budget rule matter for interpreting the results and are stated here rather than left implicit. First, the $\max(1, \cdot)$ floor means every budget retains at least one feature, so on the smallest feature sets the most aggressive budgets coincide: on Haberman ($p = 3$) the 33 % and 25 % budgets are the *same* one-feature model, and on mammographic mass ($p = 5$) the 25 % budget is a one-feature model. Second, round-half-to-even makes, for example, $p = 5$ at 50 % retain two features rather than three. Because the rule links the retained count k mechanically to p at every fractional budget, the cross-dataset analysis in Section 4.7 examines harm against the *absolute* retained count, and the exact- k budget arm evaluates every dataset at matched $k \in \{1, 2, 4, 8\}$ so that the two explanations—“few candidate features” versus “few retained features”—can be separated. Out-of-fold probabilities are pooled once per budget cell and scored. Because imputation, ranking, scaling, and model fitting all occur inside the training fold, no information from held-out patients can leak into preprocessing. The fraction of retained features is swept over $\{1.0, 0.75, 0.5, 0.33, 0.25\}$, and *the same per-repetition cross-validation split is shared across all reduction levels*, so every reduced model is compared to its own full-feature baseline as a paired observation. Algorithm 1 states this per-configuration procedure precisely.

3.3 Outcomes

For each cell we evaluate the following quantities, all of them from the pooled out-of-fold probabilities (no extra model fits): **discrimination**—AUROC; **calibration**—15-bin expected calibration error (ECE)⁶ and the Brier score⁴ (lower is better for both); **selective reliability**—AURC, the area under the risk–coverage curve with confidence taken as $|p - 0.5|$ (lower is better)^{9,10}; **clinical utility**—the standardised decision-curve net benefit $\text{NB}(t) = \frac{\text{TP}}{n} - \frac{\text{FP}}{n} \frac{t}{1-t}$ averaged over the threshold range $t \in [0.05, 0.5]$ (higher is better)¹⁹; **distribution-free reliability**—split-conformal prediction sets at 90 % nominal coverage, summarised by the mean set size (efficiency; lower is better); marginal coverage is approximately nominal (exactly so under exchangeability of the pooled out-of-fold scores), and because we compare the reduced and full-feature models built by the identical construction, our reported quantity is the *paired* set-size difference, in which any departure from exchangeability is shared and largely cancels^{20,21}; **selective calibration**—ECE on the 80 % most-confident subset; and **subgroup gaps**—the between-group range (max minus min) of AUROC, ECE, net benefit, and conformal coverage across the protected attribute’s groups (computed only where each subgroup of the protected attribute contains at least ten patients and both outcome classes; undefined, and reported as such, for the three datasets without a protected attribute). The primary estimand is, for each outcome, the mean over repetitions of the *paired difference* between the reduced-budget and the full-feature model; a positive

Algorithm 1 CERTIFYFEATUREREDUCTION: leakage-safe per-configuration evaluation.

Require: dataset (X, y, s) with p features; ranker m ; learner f ; budgets $\mathcal{F}$ (fractional and exact- k levels); repetitions R ; folds K ; master seed σ ; recalibration mode $c \in \{\text{NONE}, \text{PLATT}, \text{ISOTONIC}\}$

Ensure: metric tensor $M[\text{frac}, \text{axis}, r]$ over budgets, reliability axes, repetitions

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1: for  $r \leftarrow 1$  to  $R$  do
2:    $\rho \leftarrow \text{DERIVSEED}(\sigma, m, f, r)$  ▷ seed shared across budgets  $\Rightarrow$  paired
3:    $\{(\text{tr}_\kappa, \text{te}_\kappa)\}_{\kappa=1}^K \leftarrow \text{STRATIFIEDKFOLD}(X, y, K, \rho)$ 
4:   for each fold  $\kappa$  do ▷ training rows only: the leakage firewall
5:      $\mu_\kappa \leftarrow \text{median}(X_{\text{tr}_\kappa}); \tilde{X} \leftarrow \text{IMPUTE}(X, \mu_\kappa); \pi_\kappa \leftarrow \text{RANK}_m(\tilde{X}_{\text{tr}_\kappa}, y_{\text{tr}_\kappa})$ 
6:   end for
7:   for each budget  $\text{frac} \in \mathcal{F}$  do
8:      $k \leftarrow \max(1, \text{round}(\text{frac} \cdot p))$  ▷ exact- $k$  levels supply  $k$  directly; duplicate  $k$  dropped
9:      $\hat{p} \leftarrow$  undefined vector of length  $n$ 
10:    for each fold  $\kappa$  do
11:       $S \leftarrow \pi_\kappa[1:k]$  ▷ top- $k$  ranked features
12:       $g \leftarrow \text{FIT}(f, c, \text{STD}(\tilde{X}_{\text{tr}_\kappa, S}), y_{\text{tr}_\kappa})$  ▷  $c=\text{PLATT}$ : inner-CV recalibration
13:       $\hat{p}_{\text{te}_\kappa} \leftarrow g(\text{STD}(\tilde{X}_{\text{te}_\kappa, S}))$  ▷ standardisation fitted on  $\text{tr}_\kappa$ 
14:    end for
15:     $M[\text{frac}, \cdot, r] \leftarrow (\text{AUROC}, \text{ECE}, \text{BRIER}, \text{AURC}, \Delta_{\text{AUROC}}^s, \Delta_{\text{ECE}}^s)(y, \hat{p}, s)$ 
16:  end for
17: end for
18: return  $M$ 

```

ECE/Brier/AURC/set-size/gap difference, or a negative AUROC/net-benefit difference, means reduction *degrades* that property.

3.4 Statistics

For each (dataset, method, learner, outcome) combination, the reduced budgets form one multiplicity *family*: paired differences versus the full-feature baseline are tested with a two-sided Wilcoxon signed-rank test and corrected within the family by the Holm step-down procedure²² (R3). The fractional budgets and the exact- k budgets (Section 3) form *separate* families, so adding the exact- k arm leaves the fractional-budget corrections structurally identical to the original design. We use the non-parametric Wilcoxon test, rather than a paired t -test, because the per-cell paired differences across heterogeneous datasets are not assumed to be normally distributed and the test is robust to outliers; the alpha level is 0.05 throughout. Each difference carries a 5000-resample percentile bootstrap confidence interval (R12). A difference is called *material* only if its Holm-corrected $p < 0.05$ and its bootstrap CI excludes zero, and is then resolved by the sign of the paired difference into materially *worse* or materially *better* (Algorithm 2). Holm correction thus controls the family-wise error rate *within* each (dataset, method, learner, outcome) family across the four reduced budgets; the aggregate worse/better cell counts we report across the 108 cells are descriptive summaries of these per-family verdicts, not a single study-wide family-wise claim, and we read them as a consistent pattern of degradation rather than as one global hypothesis test. Headline numbers use 30 repetitions per cell (R4).

3.5 Pre-registration and reproducibility

Outcomes, reduction levels, and the materiality rule were set, before the data were inspected, in a deposited pre-registration. The study follows a fixed set of reproducibility rules: a number is reported only when it regenerates end-to-end from committed code, the calibration file, and a recorded seed (R1); all tables derive from a single canonical aggregator (R2); each cell uses a deterministic per-cell seed (R5); the SHA-256 of the calibration file is embedded in every results CSV file and downstream analysis refuses to mix mismatched hashes (R6); raw per-trial outputs are released in summary form with validated/provisional status tags (N14).

4 Results

All numbers below are the paired difference of an outcome at a reduced feature budget versus the full-feature model, averaged over the twelve datasets, three rankers, and three learners unless stated otherwise; a cell is one (dataset, ranker, learner) combination—108 in total—and is *material* when its Holm-corrected $p < 0.05$ and its 5000-bootstrap CI excludes zero.

Algorithm 2 AGGREGATEANDDECIDE: direction-resolved materiality.

Require: metrics M (Alg. 1) for all configurations; reference budget $\text{frac}_0=1.0$; level $\alpha=0.05$; resamples $B=5000$; per-axis degradation sign $\delta_o \in \{+1, -1\}$

Ensure: verdict $\in \{\text{WORSE}, \text{BETTER}, \text{N.S.}\}$ per (configuration, budget, axis)

```
1: for each configuration and axis  $o$  do
2:    $P \leftarrow \langle \rangle$  ▷ ordered raw  $p$ -values for the Holm family
3:   for each budget  $\text{frac} \neq \text{frac}_0$  do
4:      $d \leftarrow M[\text{frac}, o, \cdot] - M[\text{frac}_0, o, \cdot]$  ▷ paired across repetitions
5:      $p_{\text{raw}} \leftarrow \text{WILCOXON}(d)$ ;  $[\ell, u] \leftarrow \text{BOOTSTRAPCI}(d, B)$ ; append  $p_{\text{raw}}$  to  $P$ 
6:   end for
7:    $\{p_{\text{holm}}\} \leftarrow \text{HOLMSTEPDOWN}(P)$ 
8:   for each budget  $\text{frac} \neq \text{frac}_0$  do
9:      $\text{material} \leftarrow (p_{\text{holm}} < \alpha) \wedge (0 \notin [\ell, u])$ 
10:    if  $\text{material} \wedge (\text{sign}(\vec{d}) = \delta_o)$  then verdict  $\leftarrow \text{WORSE}$ 
11:    else if  $\text{material}$  then verdict  $\leftarrow \text{BETTER}$ 
12:    else verdict  $\leftarrow \text{N.S.}$ 
13:    end if
14:  end for
15: end for
```

Table 2. Paired difference at 25 % of features versus the full-feature model (mean over the twelve datasets, three rankers, and three learners; positive ECE/Brier/AURC/gap values indicate degradation, positive AUROC/net-benefit values indicate improvement). The two right columns split the Holm-significant cells (bootstrap CI excluding zero) by the direction of the difference into materially *worse* and materially *better*; for the subgroup gaps, a widening counts as worse. Computed from the released per-configuration result summary.

Outcome	Mean diff. vs. full	Material worse	Material better
AUROC ($\uparrow$ better)	-0.0398	91/108	6/108
ECE ($\downarrow$ better)	+0.0020	40/108	44/108
Brier ($\downarrow$ better)	+0.0124	78/108	19/108
AURC ($\downarrow$ better)	+0.0317	94/108	9/108
AUROC subgroup gap ($\downarrow$ better)	-0.0047	15/72 (widened)	26/72
ECE subgroup gap ($\downarrow$ better)	+0.0004	27/72 (widened)	17/72

4.1 Degradation is dose-dependent

Table 2 reports the most aggressive budget (25 % of features). At that level, selective reliability, discrimination, clinical net benefit, and conformal informativeness are materially worse than the full-feature model in the large majority of the 108 cells (AURC in 94, AUROC in 91, decision-curve net benefit in 80, conformal set size larger in 79, and the Brier score in 78), while probability calibration is mixed (ECE materially worse in 40, better in 44). For selective reliability, discrimination, and net benefit, the *mean* penalty grows monotonically as features are removed (middle and right panels of Fig. 2); the per-dataset ECE trajectories (left panel) are not monotone, consistent with the mixed calibration verdicts. The mean AURC penalty rises from +0.003 at 75 % of features (materially worse in 54 of 108 cells) to +0.032 at 25 % (94 of 108), and the AUROC drop from -0.003 to -0.040 (50 then 91 of 108), with net benefit tracking the same trajectory (-0.001 to -0.013).

4.2 Calibration and selective reliability do not merely track discrimination

A natural hypothesis is that the selective-reliability and net-benefit loss is attributable to the discrimination loss alone. Three observations argue against this. First, the reliability loss matches or exceeds the discrimination loss in extent: AURC is materially worse in 94 of 108 cells and net benefit in 80, against 91 for AUROC, with the Brier score comparable (78); the degradation thus extends beyond ranking. Second, the axes decouple: at every budget there are cells in which calibration, the Brier score, or selective reliability is materially worse while AUROC is not—13, 9, 10, and 5 of 108 at the 75, 50, 33, and 25 % budgets respectively—so these silent degradations are infrequent, and concentrated at the mildest budget, which is exactly the regime in which reduction is applied most readily and AUROC itself rarely changes materially. Third, probability calibration error behaves differently from the rest: across the twelve datasets ECE is mixed (materially worse in 40 cells but better in 44 at the most aggressive budget), whereas selective reliability, net benefit, and the Brier score degrade consistently. On several

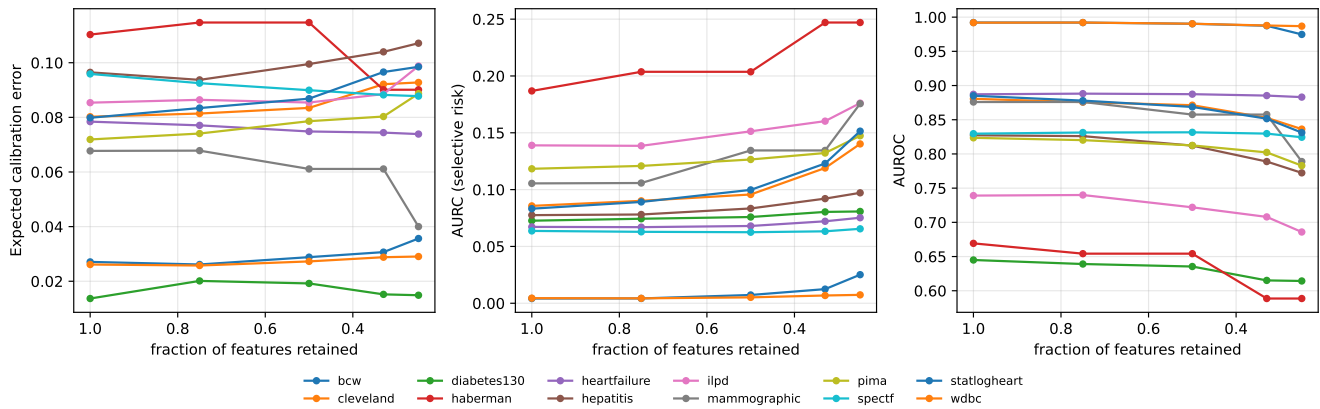


Figure 2. Calibration (ECE), selective reliability (AUC), and discrimination (AUROC) as a function of the fraction of features retained. The horizontal axis is reversed, so more aggressive reduction is to the right; one line per dataset, averaged over rankers and learners. Degradation emerges earliest and most steeply on the low-dimensional datasets (mammographic mass, Haberman, Statlog and Cleveland heart), whereas the wide-feature datasets (WDBC, SPECTF) and the large readmission cohort show minimal change—a pattern quantified by the meta-analysis (Section 4.7).

Table 3. Mean paired difference versus the full-feature model at 25 % of features, per dataset (averaged over rankers and learners; positive ECE/Brier/AURC values indicate degradation, as do negative AUROC values). Computed from the released per-configuration result summary.

Dataset	AUROC	ECE	Brier	AURC
bcw	-0.017	+0.009	+0.016	+0.021
cleveland	-0.044	+0.013	+0.023	+0.055
diabetes130	-0.031	+0.001	+0.003	+0.008
haberman	-0.081	-0.020	+0.002	+0.060
heartfailure	-0.004	-0.005	-0.003	+0.008
hepatitis	-0.055	+0.011	+0.017	+0.020
ilpd	-0.053	+0.013	+0.016	+0.037
mammographic	-0.087	-0.028	+0.021	+0.070
pima	-0.041	+0.017	+0.016	+0.029
spectf	-0.005	-0.008	+0.000	+0.002
statlogheart	-0.054	+0.019	+0.028	+0.068
wdbc	-0.005	+0.003	+0.010	+0.003

datasets aggressive reduction even *improves* the calibration error while degrading the model’s selective reliability and clinical net benefit—so calibration error and the deployment-relevant reliability of the model’s own confidence are distinct properties, and an audit that checks only one can miss the other.

4.3 The harm varies substantially across datasets

The harm is far from uniform (Fig. 2, Table 3). At 25 % of features the AURC penalty spans more than a thirty-fold range, from +0.070 on mammographic mass and +0.068 on Statlog heart down to +0.002 on SPECTF and +0.003 on WDBC. The datasets that appear robust under fractional budgets are the wide ones (WDBC, 30; SPECTF, 44; the readmission cohort, 16), where a quarter of the features is still an absolutely large set; the harmed datasets are the narrow ones (mammographic, 5; Haberman, 3; Statlog and Cleveland, 12), where the same fraction leaves a handful of features, each carrying largely unique signal that reduction cannot afford to discard. The pattern is therefore conditional: aggressive reduction is comparatively benign when the retained budget still spans the predictive signal and harmful when it does not—a relationship Section 4.7 makes quantitative, and then qualifies: the exact- k arm shows the apparent robustness of the wide datasets is a property of the *retained count*, not of width itself.

Table 4. The five new reliability axes at 25 % of features versus the full-feature model (mean over twelve datasets, three rankers, three learners; the gaps are over the nine group-bearing datasets—the ECE, AUROC, and net-benefit gaps over the eight with a valid two-class subgroup, 72 cells, and the conformal-coverage gap over all nine, 81 cells). Net benefit is a clinical-utility gain ($\uparrow$ better); the others are costs ($\downarrow$ better). Computed from the released per-configuration result summary.

Reliability axis	Mean diff. vs. full	Material worse	Material better
Net benefit (DCA) ($\uparrow$ better)	-0.0125	80/108	14/108
Net-benefit parity gap ($\downarrow$ better)	-0.0033	12/72	33/72
Conformal set size ($\downarrow$ better)	+0.0699	79/108	10/108
Group coverage gap ($\downarrow$ better)	+0.0062	26/81	10/81
Selective ECE ($\downarrow$ better)	+0.0023	44/108	45/108

4.4 Fragility is learner-dependent

The fragility is not the same across the three learners. At 25 % of features the random forest is the most fragile on every axis (AURC +0.048, ECE +0.022, net benefit -0.019, AUROC -0.053); logistic regression is the most robust on selective reliability (AURC +0.019); and gradient boosting is intermediate on reliability (AURC +0.028) but distinctive on calibration—its ECE actually *improves* under reduction (-0.018). This learner-level split is what makes the pooled ECE result mixed: the boosted model’s already-sharp probabilities tend to tighten as redundant features are dropped, while the forest’s loosen. The random-forest comparison also carries a confound, since forests produce less well-calibrated probabilities out of the box, inflating their apparent ECE penalty. The recalibration control in Section 4.9 disentangles it, showing that most of the forest’s extra calibration-error penalty is an artifact of its uncalibrated probability scale—while the selective-reliability penalty, shared by all three learners, is not.

4.5 Subgroup gaps: a dataset-specific, not systematic, fairness signal

Across the nine datasets that carry a protected attribute, the subgroup gaps do *not* move in a single direction. At the most aggressive budget the mean change in the calibration-fairness gap (ECE gap) is close to zero (+0.000), with material moves in *both* directions—widening in 27 of 72 cells and narrowing in 17; the AUROC subgroup gap leans the other way (-0.005 mean; 15 widen, 26 narrow), as does the net-benefit parity gap²³ (widening in 12 of 72 group-bearing cells, narrowing in 33), while the group-conditional conformal coverage gap widens more often than it narrows (26 of 81 versus 10). The axes therefore disagree with one another as well as across datasets—the calibration- and coverage-side gaps widen more often, the discrimination- and utility-side gaps narrow more often. The honest reading is that aggressive reduction *can* widen a fairness gap—and does so materially on individual datasets (most consequentially in the race-stratified readmission cohort, where any widening is clinically salient)—but it does not do so systematically across the broader dataset set. Fairness under reduction is therefore a per-dataset audit item, not a property that can be assumed in either direction.

4.6 Reduction reduces clinical utility and conformal informativeness

Two of the new axes translate the abstract reliability loss into clinically actionable quantities, and both degrade (Table 4). Decision-curve net benefit¹⁹—the standardised benefit of acting on the score, averaged over a clinically relevant threshold range—is materially lower at 25 % of features in 80 of 108 cells (mean -0.013), so a reduced model with preserved AUROC nonetheless yields fewer correct treatment or non-treatment decisions per patient. Split-conformal prediction sets^{20,21}, calibrated to 90 % coverage from the out-of-fold scores, become less informative: the mean set size grows in 79 of 108 cells (+0.070)—the reduced model must more often return the uninformative two-label set to keep its coverage guarantee. Because marginal coverage is held at the nominal 90 % by construction, set size, not coverage, is the informative axis. Selective calibration—ECE on the 80 % of patients the model is most confident about—is mixed (worse in 44 cells, better in 45), echoing the pooled ECE result: in the high-confidence region on which clinicians most rely, aggressive reduction improves and degrades calibration with approximately equal frequency, even as it consistently reduces net benefit and conformal informativeness.

4.7 What predicts the harm: the absolute retained count, not dataset width

The harm varies by more than thirty-fold across datasets, and that variation is systematic. The budget rule itself, however, confounds the obvious reading of it: $k = \max(1, \text{round}(\text{frac} \cdot p))$ links the retained count k mechanically to the feature count p at every fractional budget. A correlation with p therefore cannot by itself distinguish “datasets with more candidate features are safer to reduce” from “retaining very few features is unsafe”. Table 5 makes the entanglement explicit: the same 25 % budget retains one feature on mammographic mass and eleven on SPECTF, and the two datasets on which the floor binds are the two most harmed. We therefore relate the AURC penalty at the most aggressive budget to both quantities (Table 7, Fig. 3),

and then separate them with the exact- k arm. Across datasets, harm correlates with the feature count ($\rho = -0.71$, $p = 0.010$) but *more strongly with the absolute retained count* k at that budget ($\rho = -0.78$, $p = 0.003$); excluding the two datasets the rule collapses to a single feature, the p -association attenuates and is no longer significant ($\rho = -0.59$, $p = 0.075$, $n = 10$), while the retained-count association is what survives. Both correlations are robust to leaving any single dataset out (all twelve leave-one-out ρ negative; retained-count range -0.88 to -0.71) and their 5000-resample bootstrap intervals exclude zero ($[-0.98, -0.32]$ for the retained count). The intrinsic dimensionality is a corroborating predictor ($\rho = +0.60$, $p = 0.037$); the sample-to-feature ratio, feature redundancy, and outcome prevalence are not significant. With twelve datasets this cross-dataset layer remains exploratory.

The exact- k arm separates the explanations. Evaluated at matched retained counts $k \in \{1, 2, 4, 8\}$ (each inside the same repetitions and folds as the fractional arm, in its own multiplicity family), *every* dataset is materially harmed at $k = 1$ —including the wide datasets whose fractional budgets never approach it: SPECTF ($p = 44$), whose own 25 % budget retains eleven features at a penalty of $+0.002$, pays $+0.035$ at $k = 1$; WDBC $+0.030$; the full readmission cohort $+0.027$; and the compact heart datasets pay an order of magnitude more (Statlog $+0.191$, Cleveland $+0.178$). The apparent robustness of wide datasets is therefore a property of the *retained count*—their fractional budgets simply never visit the damaging regime—while dataset structure still modulates the severity at matched k by an order of magnitude. Two further analyses complete the picture. First, the harm is not attributable to instability of the selection itself. We measured the fold-to-fold stability of the selected top- k sets (mean pairwise Jaccard over all 150 per-fold rankings per dataset and ranker; Fig. 4). The decisive evidence is a dissociation rather than a correlation: Haberman is the *second most stable* dataset in the study (Jaccard 0.93) and the *third most harmed* ($+0.060$), while SPECTF is among the least stable (0.51) and the *least harmed* ($+0.002$). Stability neither protects a dataset nor condemns it, and the exact- k arm makes the same point without reference to stability at all: at $k = 1$ every dataset is materially harmed, including those whose selections are nearly deterministic. We report the correlational test as measured, in both directions: across datasets it is null ($\rho = +0.14$, $p = 0.66$, $n = 12$), and at the finer (dataset, ranker) level it is null when pooled ($\rho = -0.047$, $n = 36$) but shows a weak within-dataset trend ($\rho = -0.29$; $p = 0.19$ by a block permutation over the twelve independent datasets) whose *sign* is the one a selection-noise mechanism would predict. That trend is not significant, and it is confounded—within a dataset, a better ranker is plausibly both more stable and less harmful, so the association need not run from instability to harm. We therefore do not rest the claim on it: the damage is the information discarded, not noise in what is kept, because datasets whose selections barely move are harmed anyway. Second, a strict certificate—requiring *no* materially-worse cell on any reliability axis at a budget—is almost never attainable: of the twelve datasets, only WDBC admits any certified-safe reduction at all (at $k = 22$ of $p = 30$; Table 6). The practical conclusion is that no proxy—neither dataset width nor any other property we measured—certifies a budget in advance. The retained count identifies the *high-risk* regime (small k , most severely below $k \approx 4$), and the audit itself is the certificate.

Because that is a negative claim, we gave one candidate proxy a genuine chance to refute it, under thresholds fixed in a deposited pre-registration before the diagnostic was computed. The candidate was the ranker’s concentration curve: define the saturation point k^* as the smallest budget at which the top- k score share reaches 90 %, and predict that any budget below k^* is unsafe. Across 80 dataset-by-budget cells the rule agrees with the measured verdict 59 % of the time. It is not a predictor but a one-sided screen: sensitivity is 1.00—it flags every budget that genuinely degrades selective reliability, with zero false negatives—while specificity is 0.15, flagging 33 budgets that turn out to be safe. A screen that says “unsafe” almost everywhere is not a certificate. Two details are worth keeping: the datasets whose curves never reach saturation within the grid are exactly the ones the probe-injection experiment identified as having distributed signal, and every cell the rule did call safe was in fact safe—but on six such cells, which is far too few to rest anything on, and we do not.

One candidate proxy deserves explicit dismissal, because it is the one a reader is most likely to suspect: *cohort size*. Across the twelve datasets the association between n and the harm is absent to begin with (Spearman $\rho = +0.05$ against the AUROC penalty, $\rho = +0.22$ against AURC, neither remotely significant), and what little of it exists disappears once p is held fixed (partial $\rho = -0.26$ and -0.20). By contrast p survives the reverse conditioning (partial $\rho = -0.71$ against AURC, $p = 0.015$) and the retained count is stronger still ($\rho = -0.78$, $p = 0.003$). Sample size, in other words, was p wearing a disguise. We report this because the intuition that small cohorts are the fragile ones is natural, widely held, and—on these datasets—not what the measurements say. With twelve datasets these correlations are descriptive rather than confirmatory, and we treat them as such.

The retained count is not the whole account either, and the same collinearity that misled us about sample size applies to it: because the fractional budget sets $k = \lceil 0.25p \rceil$, a small retained count and a narrow dataset are hard to tell apart. We therefore repeated the comparison at a *matched absolute* budget, at every exact- k level the design provides. A dataset enters at budget k only when $p > k$, so that reduction is genuine; at $k = 1$ and $k = 2$ all twelve datasets qualify, and the panel narrows to eleven and nine at $k = 4$ and $k = 8$.

The width association is present on the full panel and strengthens as the budget loosens: Spearman $\rho = +0.32$ ($p = 0.31$) at $k = 1$, $+0.60$ ($p = 0.039$) at $k = 2$, $+0.69$ ($p = 0.020$) at $k = 4$ and $+0.77$ ($p = 0.016$) at $k = 8$. From $k = 2$ upward it is

Table 5. What the fractional budgets actually retain. For each dataset: the candidate count p , the retained count k at the most aggressive budget, the AURC penalty there, its harm rank, and whether the $\max(1, \cdot)$ floor is what produced that k . The two floor-collapsed datasets are the two most harmed, which is why the harm must be related to k rather than to p (Section 4.7); the exact- k arm removes the confound entirely by evaluating every dataset at the same k . Computed from the released per-configuration result summary.

Dataset	p	$k@25\%$	harm	ρ -rank	floor?
mammographic	5	1	+0.070	1	✓
statlogheart	12	3	+0.068	2	–
haberman	3	1	+0.060	3	✓
cleveland	12	3	+0.055	4	–
ilpd	9	2	+0.037	5	–
pima	8	2	+0.029	6	–
bcw	9	2	+0.021	7	–
hepatitis	18	4	+0.020	8	–
diabetes130	16	4	+0.008	9	–
heartfailure	11	3	+0.008	10	–
wdbc	30	8	+0.003	11	–
spectf	44	11	+0.002	12	–

significant on all twelve datasets—including the three narrowest, which the widest budget cannot assess at all. So the association with retained count reported above is partly the collinearity, and what a fixed budget *discards*—which grows with p —matters independently of what it keeps.

That the effect vanishes at $k = 1$ is not a failure of the pattern but a consequence of it. Reducing every dataset to a single feature harms all of them, wide and compact alike, as the exact- k arm reports directly; with no dataset spared there is nothing left for width to discriminate. The two quantities are therefore not rivals, and the practical conclusion is unchanged, because neither certifies a budget in advance.

4.8 Beyond the audited regime: what governs reduction when $p \gg n$

The analysis above is scoped to low- and moderate-dimensional tabular data ($p = 3\text{--}44$). Because feature reduction is most often applied where features are many, we ran a separate, exploratory arm on five high-dimensional biomedical datasets to ask whether the same account holds there. It uses an *absolute- k* budget grid rather than fractional budgets—at $p = 22,283$ a 25 % budget retains more features than there are patients, so fractional budgets never reach the regime in which the harm appears—and it is reported as a separate artifact under its own configuration, never pooled with the twelve clinical datasets.

The result is a negative one for the obvious hypothesis. Ordered by dimensionality, the harm from aggressive reduction is *not* monotone: at $p = 279$ (arrhythmia, $n = 452$) the AUROC penalty at $k = 1$ is -0.278 ; at $p = 5,966$ (prostate, $n = 102$) it is only -0.043 and reduction is neutral-to-beneficial beyond $k = 8$; at $p = 10,000$ (arcene, $n = 200$) it returns to -0.298 , the most harmful of the five; at $p = 19,993$ (smokers’ airway epithelium, $n = 187$) it falls again to -0.176 ; and at $p = 22,283$ (glioma, $n = 85$) it is -0.159 . The sequence rises, falls, rises and falls again across two orders of magnitude in width. **Higher dimensionality does not predict safer reduction.**

Nor is the ordering explained by sample scarcity relative to features. Prostate and arcene have almost identical n/p (0.017 and 0.020) and opposite behaviour—the first tolerating aggressive reduction, the second the worst-harmed dataset we measured. We tested this directly rather than inferring it: holding $p = 279$ fixed on arrhythmia and subsampling n to 100, reaching $n/p \approx 0.36$, leaves the penalty essentially unchanged (random forest -0.105 against -0.099 at full n), and the variation across n is small and non-monotone. Moving n alone does not move the harm.

What does separate the extremes is how much of the ranker’s score mass falls inside the budget. The quantity that matters for an absolute- k budget is not scale-free inequality across all features but the share of score mass in the top k , since that is precisely what a k -feature model retains. Between the two datasets whose widths are comparable—prostate ($p = 5,966$) and arcene ($p = 10,000$)—the contrast is unambiguous and holds for *every* ranker and every concentration statistic we computed. Random-forest importance places 38 % of its score mass in prostate’s top 64 features against 14 % for arcene, and L_1 -logistic places 66 % in prostate’s top 8 against 15 %; mutual information gives 6.1 % against 2.8 % at the same budgets. Where the signal is concentrated within the budget, a small budget preserves it; where it is diffuse, the same budget discards it. Arcene is the instructive case: it is constructed with a large number of noise features, and although its scores are unequal in the scale-free sense (Gini 0.83 under random-forest importance), its top 8 features carry only 2.9 % of the mass under that same ranker,

Table 6. The strict safety certificate almost never passes. For each dataset, the smallest retained count k at which *no* (ranker, learner) cell is materially worse than the full-feature model on *any* reliability axis (ECE, Brier, AURC, net benefit, conformal set size, AUROC), probing every budget of both arms. Computed from the released per-configuration result summary.

Dataset	p	smallest certified-safe k	as fraction of p
wdbc	30	22	0.73
bcw	9	<i>none below full</i>	—
cleveland	12	<i>none below full</i>	—
diabetes130	16	<i>none below full</i>	—
haberman	3	<i>none below full</i>	—
heartfailure	11	<i>none below full</i>	—
hepatitis	18	<i>none below full</i>	—
ilpd	9	<i>none below full</i>	—
mammographic	5	<i>none below full</i>	—
pima	8	<i>none below full</i>	—
spectf	44	<i>none below full</i>	—
statlogheart	12	<i>none below full</i>	—

Table 7. Dataset properties and the reliability harm (AURC penalty at 25 % of features), sorted by harm. The harm is predicted by the feature count (p) and the intrinsic-dim fraction (see text). Computed from the released per-configuration result summary and the dataset definitions.

Dataset	p	$n:p$	Redund.	Intr.dim	Prev.	AURC penalty
mammographic	5	192.20	0.21	1.00	0.46	+0.070
statlogheart	12	22.50	0.17	0.92	0.44	+0.068
haberman	3	102.00	0.05	1.00	0.26	+0.060
cleveland	12	25.25	0.17	0.92	0.46	+0.055
ilpd	9	64.78	0.21	0.67	0.71	+0.037
pima	8	96.00	0.17	0.88	0.35	+0.029
bcw	9	77.67	0.60	0.78	0.34	+0.021
hepatitis	18	8.61	0.18	0.89	0.21	+0.020
diabetes130	16	6360.19	0.09	0.88	0.11	+0.008
heartfailure	11	27.18	0.07	1.00	0.32	+0.008
wdbc	30	18.97	0.39	0.33	0.37	+0.003
spectf	44	6.07	0.29	0.57	0.79	+0.002

so a $k = 8$ model discards almost all of it. That reconciliation—high inequality, low in-budget share—is exactly why the budget-relevant statistic is the one that tracks the harm.

Concentration does *not* order these datasets. Under L_1 -logistic ranking the four do fall in the predicted order at every budget we examined ($k \in \{1, 2, 4, 8\}$), which is suggestive. It does not replicate. Under mutual information and random-forest importance the order fails at every budget, and it fails the same way each time: arrhythmia ranks as the most concentrated of them, which is an artefact of its width—at $p = 279$ a top-64 share covers 23 % of its features against 0.6 % of arcene’s. The scale-free alternative disagrees in a third direction, ranking glioma first and arcene above arrhythmia under all three rankers.

The incomparability has an obvious correction, so we applied it: divide each top- k share by k/p , the share a uniform ranker would achieve, giving the lift of the observed concentration over chance at the same width. Computed for every ranker at every budget across all five datasets—45 combinations of statistic, ranker and correction, all reported—the ordering is reproduced zero times, by the raw statistic and the corrected one alike, and the correction places arrhythmia *last* in concentration although it is the second-most harmed. This is a negative result about the explanation proposed above.

What does survive is narrower and robust. Prostate is more concentrated than arcene, and less harmed, under every ranker, every budget and every statistic including the width-corrected one. The concentration account therefore separates the extremes of this arm, and we do not extend it to a complete ordering, because the measurements do not support one.

A cross-dataset correlation cannot establish a mechanism in any case, so we ran a controlled, within-dataset test of it and pre-registered the prediction before computing. If diffuse signal is what makes reduction harmful, then *diluting* a dataset—

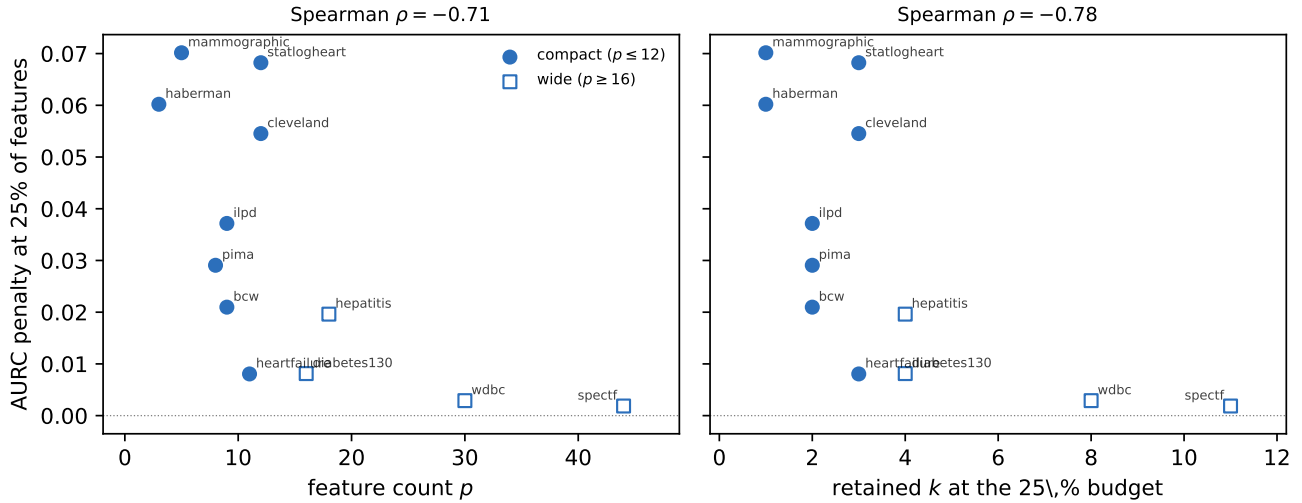


Figure 3. Reliability harm (AUCR penalty at 25 % of features) versus the feature count p (left) and versus the absolute retained count k at that budget (right), one point per dataset; filled circles mark compact datasets ($p \leq 12$), open squares wide ones ($p \geq 16$). The retained count is the stronger correlate, and the exact- k arm (text) shows that at matched $k = 1$ every dataset, wide or compact, is materially harmed.

adding pure-noise features while holding the rows, the true features, the folds and the *absolute* budget fixed—should increase the harm. We injected Gaussian probes drawn independently of the outcome at $\{1, 2, 5, 10\} \times$ the original feature count into three datasets, holding the reduced model at the same absolute k throughout ($k = 3$ for cleveland, 1 for mammographic, 11 for spectf), so that the only quantity changing is the number of irrelevant features.

The prediction was wrong, and the correction is informative. Dilution did not raise the penalty; it lowered it. On cleveland the AUROC reduction penalty fell from $+0.054$ (95 % CI $+0.050$ to $+0.058$) undiluted to $+0.006$ (-0.002 to $+0.013$) at ten-fold padding: the intervals are disjoint and the diluted one covers zero, so the harm is not merely reduced but abolished. On spectf the penalty crossed from $+0.004$ ($+0.001$ to $+0.007$) to -0.037 (-0.042 to -0.032), a sign reversal with no overlap—reduction becomes actively beneficial. On mammographic it did not move ($+0.086$ to $+0.081$, intervals overlapping) and stayed materially positive at every dose. Each cell aggregates 180 units (30 repetitions $\times 3$ rankers $\times 2$ learners). The reason is visible in the two models separately: the reduced model is largely indifferent to the injected noise, because the ranker keeps recovering the true features, while the full model degrades as the noise accumulates—so the gap between them closes. *Reduction defends against noise padding.* Harm from aggressive reduction is therefore not caused by a dataset merely containing many irrelevant features, which is the intuition the high-dimensional arm was designed to test and which these data refute.

That refinement sharpens the account rather than dissolving it, and it explains the null above. What distinguishes arcene from prostate is not that arcene has more noise, but that its predictive signal is genuinely *spread across many features*: reducing to a handful discards real signal, whereas prostate’s signal survives the same budget because fewer features carry it. The operative quantity is the number of features bearing signal—an effective dimensionality—and that is precisely what a single scalar concentration index cannot express across datasets differing in width by two orders of magnitude. A cross-dataset index gave no ordering; a within-dataset intervention gave a mechanism.

A second arm tests the same mechanism from the opposite direction. Rather than adding noise, restrict arcene to its top- N features and reduce *within* that restricted set at a fixed absolute budget of esult25, so that the budget is constant and only the composition of the available features changes. The kept set is selected inside each training fold, like every other selection in this study, and the per-cell seed excludes N , so every rung of the ladder sees the identical cross-validation splits and the identical base ranking—the rungs differ only in how deep that one ranking is cut, and each is paired with every other.

The penalty rises monotonically as the available set widens, from esult $+0.025$ (95 % CI esult $+0.022$ to esult $+0.028$) when only the top esult100 features are available to esult $+0.091$ (esult $+0.082$ to esult $+0.100$) with all esult10,000. Because the rungs are paired, the comparison can be made within units rather than between them: *every* adjacent step is positive with an interval excluding zero (esult $+0.010$, esult $+0.024$, esult $+0.017$, esult $+0.014$), and the end-to-end paired difference is esult $+0.066$ (esult $+0.056$ to esult $+0.076$) over esult180

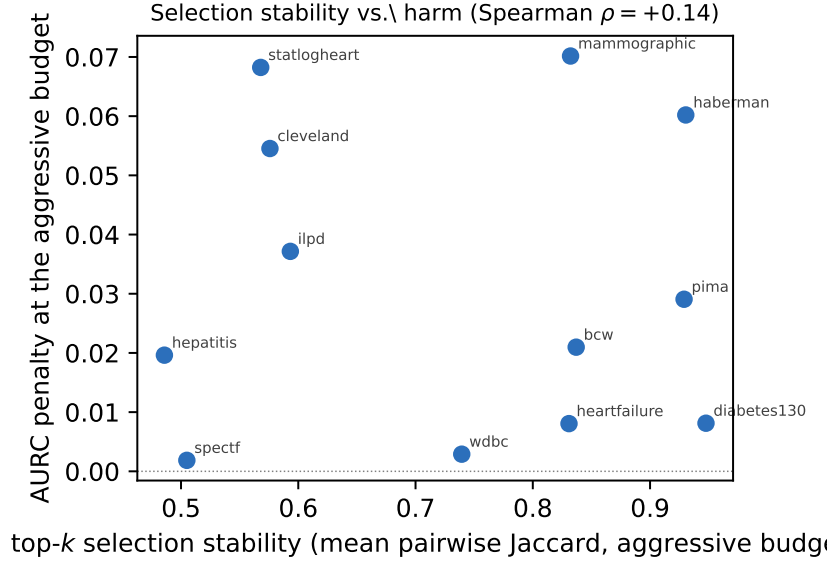


Figure 4. The harm is not selection noise: the fold-to-fold stability of the selected top- k feature sets (mean pairwise Jaccard across all per-fold rankings) does not predict the AUCR penalty across datasets (Spearman $\rho = +0.14$, $p = 0.66$, $n = 12$). The dissociation is what carries the point—Haberman is nearly the most stable dataset and among the most harmed, SPECTF among the least stable and the least harmed. A finer (dataset, ranker) analysis is reported in the text, including a weak, non-significant within-dataset trend in the direction a noise mechanism would predict.

units.

The decomposition names the cause. The full-feature reference barely moves across the ladder (AUROC result0.898 to result0.911) while the result25-feature model falls steadily (result0.873 to result0.821). A fixed budget of twenty-five features therefore captures a smaller and smaller fraction of arcene’s signal as more of that signal becomes available to lose, which is what a large effective signal dimensionality predicts and what a concentrated one does not.

We report this arm as exploratory and do not extend the paper’s guidance to this regime. Its lesson is nonetheless the same as the main analysis: no simple property of a dataset—not width, not n/p —certifies that reduction is safe, and the property that comes closest is one that must be measured on the ranking rather than read off the data’s shape.

4.9 Recalibration recovers the calibration error, but not the selective-reliability or net-benefit loss

A standard approach to a calibration problem is post-hoc recalibration, which we evaluate as a control at two depths. In a separate, leakage-safe experiment we wrapped each base learner in either Platt (sigmoid) scaling or the more flexible isotonic regression⁵, each fitted by an inner 3-fold cross-validation *within* each training fold—so the calibrator never sees the outer held-out fold—and recomputed the reduction penalty at the full and 25 % budgets (Table 9). The result is a dissociation with a learner-specific calibration component.

Where a raw calibration-error penalty exists, recalibration removes it. Pooled over all three rankers, the random forest’s ECE penalty collapses from +0.0215 to +0.0020 under Platt scaling and +0.0057 under isotonic regression, and the number of materially worse cells falls from 20 of 36 to 12. The forest’s apparent fragility is therefore largely its uncalibrated probability scale, not reduction itself. The other two learners have little to recover: the raw ECE penalty is already small for logistic regression, and for gradient boosting it is negative, meaning reduction *improves* its calibration. Pooled over all learners the ECE penalty is accordingly unchanged (+0.0020 raw against +0.0021 under Platt).

Gradient boosting is worth stating separately, because it shows the dissociation in its sharpest form. In a dedicated sweep over all twelve datasets and all three rankers, aggressive reduction *materially improves* that learner’s uncalibrated calibration error—a Holm-surviving effect at every bin count tested—while its selective-reliability penalty is simultaneously +0.028, positive and of the same magnitude as the other learners’. A model can therefore become better calibrated and less able to rank its own errors under the same intervention. That is the dissociation stated without the confound of a learner whose raw calibration was poor to begin with, and it holds for all three learners rather than two.

The deployment-relevant penalties behave differently. They *persist* under recalibration at both depths, for every learner and for every ranker. The pooled AUCR penalty moves only from +0.0317 raw to +0.0313 under Platt and +0.0304 under isotonic;

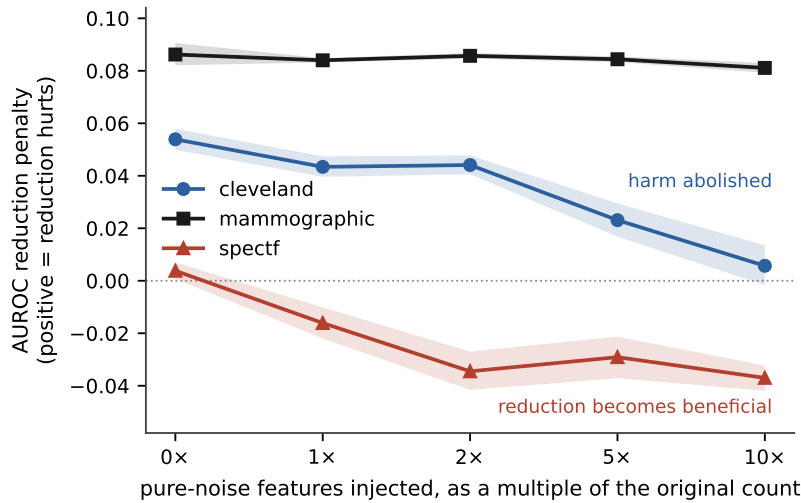


Figure 5. Reduction defends against noise padding. AUROC reduction penalty as pure-noise features drawn independently of the outcome are injected, with rows, true features, folds and the *absolute* retained budget held fixed, so that the only quantity changing is the number of irrelevant features. Bands are 95 % bootstrap intervals over 180 units per cell (30 repetitions $\times$ 3 rankers $\times$ 2 learners). The pre-registered prediction was that dilution would raise the penalty. It falls to zero on cleveland, crosses into benefit on SPECTF, and does not move on mammographic—so the harm from aggressive reduction is not caused by a dataset merely containing many irrelevant features.

the random forest’s moves from +0.0475 to +0.0411 and +0.0386. It remains materially worse in 95 to 96 of 108 cells under either recalibrator, with comparable persistence in the Brier and net-benefit penalties. Notably, isotonic regression—the more flexible recalibrator—does not recover them either; we return below to a five-rung flexibility ladder that settles whether this is a limitation of the recalibrator or a property of the metric. Indeed for one ranker recalibration makes selective reliability *worse* rather than merely failing to help: under random-forest importance the AURC penalty rises from +0.0399 raw to +0.0426 under Platt. Recalibration thus fixes the part of the problem that is a relabelling of probabilities, but not the part that is a loss of the model’s ability to rank its own errors and to deliver clinical net benefit—the property selective prediction and threshold-based decision-making depend on. This control now spans *all three* rankers rather than one: the dissociation holds for each of mutual information, random-forest importance and L_1 -logistic selection, whose raw AURC penalties (+0.0291, +0.0399 and +0.0263) move to +0.0273, +0.0426 and +0.0241 under Platt. Random-forest importance carries the largest penalties of the three rankers, so the dissociation is not an artefact of how features were ranked.

A natural objection is that two recalibrators are too thin a basis for a negative claim: perhaps some better-specified method would recover what Platt and isotonic cannot. We therefore extended the control to two further post-hoc methods on the same twelve datasets, three rankers and three learners, fitted by the same inner 3-fold procedure within each training fold—temperature scaling, a single-parameter method, and beta calibration, a three-parameter parametric family. Neither changes the picture. At the full budget, temperature scaling reduces the mean calibration error from 0.0694 to 0.0640 (−8 %) and beta calibration to 0.0504 (−28 %), while the deployment-relevant axes barely move: mean AURC changes by −3 % and +0.6 % respectively, and mean net benefit by 0.0 % and +0.8 %. Beta calibration is the decisive case, because it is a genuinely flexible *parametric* family: it cuts calibration error by more than a quarter and leaves selective reliability where it found it.

Ordering all five recalibrators by flexibility makes the point as a single object (Table 8). Recovery of calibration error rises *strictly monotonically* with flexibility, from +0.0054 at one parameter to +0.0193 for the nonparametric recalibrator (Spearman $\rho = +1.00$), reaching 27.8 % of the uncalibrated baseline. Selective reliability does not follow: AURC recovery is negative at three of the four rungs and its correlation with flexibility runs the *opposite* way ($\rho = -0.90$), with the largest move in either direction worth only 3.1 % of the baseline—about 8 % of the AURC penalty that reduction itself inflicts. Net-benefit recovery does rise monotonically with flexibility ($\rho = +1.00$), but its largest value is +0.0028, or 0.9 % of baseline: a thirtieth of what the same ladder achieves for calibration error. The failure to recover therefore cannot be attributed to an insufficiently flexible recalibrator, because the most flexible recalibrators are precisely the ones that recover the most calibration error and the least selective reliability.

We think the mechanism is straightforward, and stating it makes the dissociation less surprising and more useful. Post-hoc recalibration is a near-monotone reparametrisation of the probability *scale*; expected calibration error is defined on that scale,

Table 8. The recalibrator flexibility ladder at the full budget, over twelve datasets, three rankers and three learners (108 cells at every rung). Recovery is signed so that positive means improvement over the uncalibrated model. Calibration-error recovery rises strictly with flexibility; selective-reliability recovery does not, and is negative at most rungs. Net-benefit recovery rises monotonically but is an order of magnitude smaller, as the final row shows.

Recalibrator	Free parameters	Calibration error		Selective reliability	Net benefit
		ECE	recovery	AURC recovery	recovery
None (uncalibrated)	0	0.0694	+0.0000	+0.0000	+0.0000
Temperature scaling	1	0.0640	+0.0054	+0.0026	+0.0001
Platt (sigmoid)	2	0.0542	+0.0152	−0.0004	+0.0021
Beta calibration	3	0.0504	+0.0191	−0.0005	+0.0024
Isotonic regression	nonparametric	0.0501	+0.0193	−0.0013	+0.0028
Spearman ρ with flexibility			+1.00	−0.90	+1.00
Largest move, % of uncalibrated baseline			27.8 %	3.1 %	0.9 %

Table 9. Raw versus leakage-safe Platt- and isotonic-recalibrated reduction penalty at 25 % of features (mean paired difference vs. full over the twelve datasets, *all three rankers*, and three learners—108 cells; positive ECE/Brier/AURC values indicate degradation, as do negative AUROC/net-benefit values), with the count of materially worse cells. Recalibration recovers the (learner-specific) calibration-error penalty but not the AURC, Brier, or net-benefit loss, and isotonic does no better than Platt. The pooled ECE row is dominated by the learners with little or no raw penalty to recover; the recovery is visible in the random forest, reported in the text. Computed from the released all-rankers recalibration-control summary.

Outcome	Raw penalty (worse)	Platt penalty (worse)	Isotonic penalty (worse)
AUROC (↑)	-0.0398 (92/108)	-0.0410 (96/108)	-0.0392 (94/108)
ECE (↓)	+0.0020 (43/108)	+0.0021 (36/108)	+0.0048 (56/108)
Brier (↓)	+0.0124 (79/108)	+0.0143 (96/108)	+0.0144 (93/108)
AURC (↓)	+0.0317 (95/108)	+0.0313 (96/108)	+0.0304 (96/108)
Net benefit (↑)	-0.0125 (81/108)	-0.0142 (90/108)	-0.0137 (90/108)

which is why recalibration moves it and why more flexible maps move it further. Selective reliability is defined on the probability ordering—which cases the model is willing to stake a decision on, relative to which others. A monotone map of the scale leaves that ordering almost untouched, so no amount of recalibration flexibility can repair it.¹ What aggressive reduction damages is the ordering, and an intervention that acts only on the scale is the wrong instrument for it. All five rungs are computed on the identical 108-cell grid, and the two source runs’ uncalibrated baselines agree to every printed digit on all three axes.

4.10 The full readmission cohort, and what the subsample would have missed

We analyse all 101,763 eligible encounters of the Diabetes-130 cohort, retaining a deterministic 6,000-encounter subsample as a robustness arm (the full-feature discrimination, AUROC $\approx$ 0.65 across cells, is in the moderate range expected for readmission prediction on this cohort¹⁸). Re-running the subsample under the same protocol as a robustness arm (Table 10) shows why the full cohort matters. Point estimates of the reduction penalty agree closely in the typical case (median absolute difference 0.004); but of the 45 (ranker, learner, outcome) comparisons at the aggressive budget, materiality verdicts disagree in 20—in 12 the full cohort resolves a material effect the subsample was underpowered to see, and in 8 the subsample asserted materiality the full cohort refutes. The subsample also *reversed the sign* of the measured effect in 18 of the 45 comparisons, most consequentially on discrimination (mutual information $\times$ random forest, AUROC: +0.028 full versus −0.028 subsampled). The subsample was therefore not merely noisier: for small effects it could mislead in either direction, and all conclusions in this paper rest on the full cohort.

4.11 A worked clinical example: reduction flips real treatment decisions

Threshold-averaged net benefit quantifies clinical utility but obscures individual decisions, so we fix a clinically meaningful operating point and count them. On the mammographic-mass data the decision is whether to biopsy; a predicted malignancy

¹Almost, rather than exactly: confidence is $|p - 0.5|$, and a recalibration map with a non-zero intercept moves the point at which the two classes are equally likely, so the induced ordering can shift slightly. That is the small non-zero AURC movement in Table 8, and it is why the correlation with flexibility is −0.90 rather than undefined.

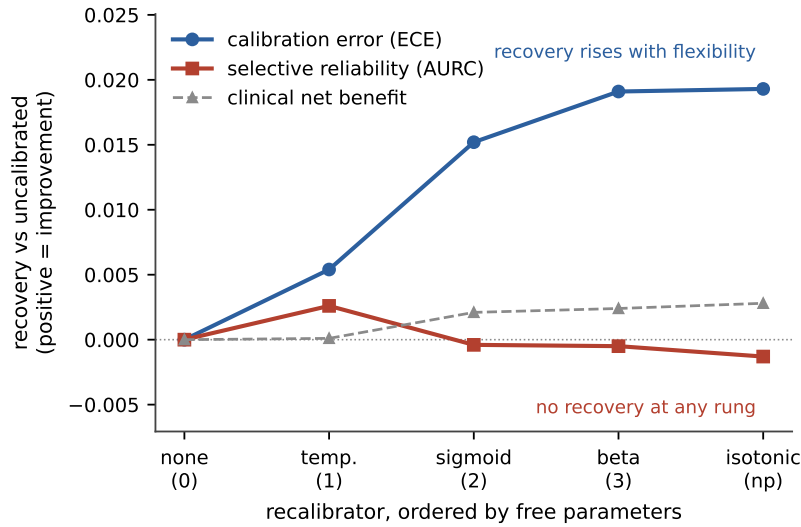


Figure 6. Post-hoc recalibration repairs the probability scale in proportion to its flexibility, and repairs the ordering at no rung. Recovery relative to the uncalibrated model at the full budget, over the identical 108-cell grid at every rung, with recalibrators ordered by free parameters. Calibration-error recovery climbs monotonically to 27.8 % of the uncalibrated baseline; selective-reliability recovery is flat-to-negative throughout, and clinical net benefit, though monotone, recovers an order of magnitude less. The gap between the two curves is the dissociation the paper reports, and it cannot be closed by choosing a more flexible recalibrator.

risk of 10 % is the conventional boundary at which biopsy is recommended, and the cell (mutual-information ranker, logistic regression, uncalibrated) was fixed *a priori* in the analysis configuration. At the 25 % budget the reduced model retains a single feature, so the comparison is precisely “the full workup versus one predictor”. The result (Table 11, Fig. 7): 203 of 961 patients (21 %) have their biopsy decision flipped in the majority of the 30 paired models, 257 in at least one; among patients whose mass is malignant, 15 flip in the majority, of whom 4 are moved *from biopsy to no biopsy*—the costly direction. The count is threshold-dependent (92 and 266 patients at $t = 0.05$ and $t = 0.15$), but not the phenomenon. At population scale, the same analysis on the full readmission cohort at $t = 0.10$ (approximately the cohort prevalence) flips the program-enrollment decision for 14,324 of 101,763 encounters (14.1 %) in the majority of paired models, including 1,532 true readmissions, of which 1,132 lose the intervention (Fig. 8). Decision instability is not concentrated on minority race groups in this cohort: mean flip rates span 0.112 (Hispanic) to 0.149 (Caucasian) (Table 12)—consistent with Section 4’s broader finding that fairness effects are dataset-specific rather than systematic, and reported here as measured. The reliability diagrams for the same cell (Fig. 9) show the mechanism: the reduced model’s probabilities are not merely shifted but compressed toward the prevalence, so threshold crossings multiply near the operating point. These decision-level counts are what the statistical materiality of Sections 4–4.7 means at the bedside.

5 Discussion

The audit demonstrates, in plain terms, that good discrimination in a reduced model is not a reliable indicator of good calibration, of dependable confidence, or of preserved clinical utility—so discrimination alone cannot certify a reduced model as safe. Across twelve public clinical datasets, three learners, and three standard rankers, reducing the feature budget was associated, at aggressive cuts, with degraded selective reliability (AURC, worse in 94 of 108 cells), reduced clinical net benefit (80 of 108), and larger, less informative conformal prediction sets (79 of 108): a loss comparable in extent to, and partly independent of, the loss of discrimination, and present in cells where discrimination remained materially preserved. That selective-reliability figure carries a measured qualification. AURC is computed from a confidence that is itself a function of the predicted probability, so part of the degradation it reports is a property of the probability scale. Recomputing the same penalty with a confidence taken from the feature space alone—distance to the training support, which the model’s probability never enters—drops it from $+0.0334$ to $+0.0116$. Roughly two-thirds of the measured selective-reliability harm is therefore scale-borne. The remaining third survives a confidence that cannot see the probability at all, decisively ($p = 7 \times 10^{-103}$) and with no dependence on cohort size, which is what makes it evidence of a genuine loss of error-ordering rather than a re-description of the calibration result. Probability calibration, by contrast, was mixed across the broader dataset set: aggressive reduction improved and degraded

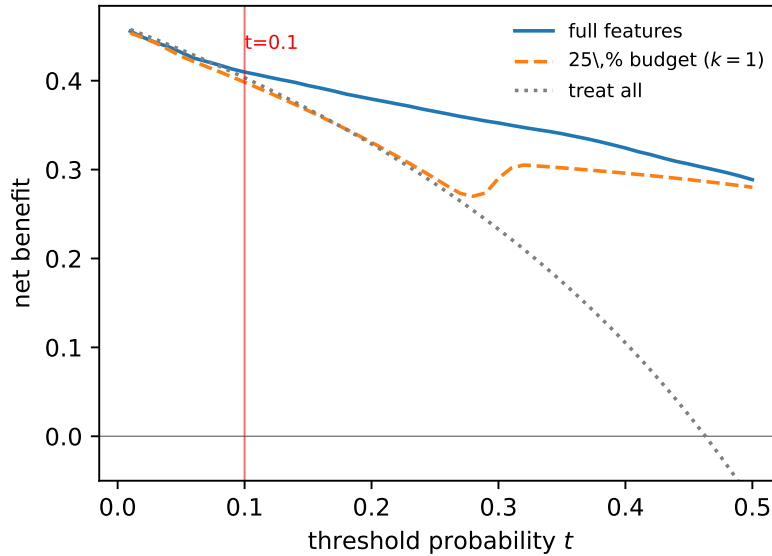


Figure 7. Per-threshold decision curves for the worked-example cell (mammographic mass), full versus one-feature reduced model, with treat-all and treat-none references; the vertical line marks the biopsy threshold $t = 0.10$. Computed from the released per-threshold net-benefit cache.

the calibration error with approximately equal frequency, driven by the gradient-boosted learner whose sharp probabilities tightened as redundant features were dropped. The deployment-relevant harms—degradation in the trustworthiness of the model’s confidence and in the net benefit of acting on it—are therefore the most robust signal, and none of them is visible to the AUROC-only evaluation by which feature reduction is usually justified.

Stating the dissociation as a correlation makes its shape clearer than the scattered observations above. Orienting every axis so that positive means worse, and correlating the reduction penalties across all 108 cells, the reliability axes do not form one scale. Calibration error and selective reliability correlate at only $\rho = +0.30$: knowing that reduction worsened a model’s calibration tells you little about whether it worsened the model’s ability to rank its own errors. But selective reliability and conformal set size correlate at $\rho = +0.89$, and calibration error correlates with selective calibration error at $\rho = +0.97$ —the latter a sanity check, since the two are the same quantity on different subsets. The pattern is not noise: the axes that depend on the *ordering* of predicted probabilities move together, the axes defined on the probability *scale* move together, and the two groups are nearly independent of each other. That is the same division the recalibration ladder implies mechanically (Section 4.9)—recalibration repairs the scale and cannot repair an ordering—arriving here from a different direction, and it is why a single calibration number cannot stand in for the audit.

Why selective reliability matters here. AURC summarises whether a model’s own confidence can be trusted to triage which predictions to act on. A reduced model with a larger AURC is one whose abstention or referral behaviour is less dependable—a property of direct operational relevance in decision support, and one that neither AUROC nor a single calibration number captures. That AURC degrades early and steeply under reduction (Fig. 2) is the most deployment-relevant property the protocol exposes.

Conditions under which reduction is safe. The extent of degradation is dataset-dependent rather than intrinsic to feature reduction; the retained-count analysis (Section 4.7) shows that a width-based screen (“more candidate features means safer reduction”) conflates two explanations the budget rule links mechanically. Separated by the exact- k arm, the statement is narrower: harm tracks the *absolute number of retained features* more strongly than dataset width ($\rho = -0.78$ versus $\rho = -0.71$, both leave-one-out robust); at matched $k = 1$ every dataset, wide or compact, is materially harmed; dataset structure modulates severity at matched k by an order of magnitude; and datasets whose selected subsets barely move across folds are harmed anyway, so instability of the selection does not explain the harm. What survives as *a priori* guidance is therefore a risk gradient, not a certificate: budgets that retain very few features ($k \lesssim 4$) are high-risk on any dataset, and no proxy we measured—width, redundancy, intrinsic dimension, sample size, prevalence, or selection stability—identifies a safe budget in advance. The strict certificate (no materially-worse cell on any axis) passed for exactly one dataset at one budget in this study. The audit itself, run per dataset, learner, and retained count, is the certificate. We do not assert that feature reduction is universally harmful; we

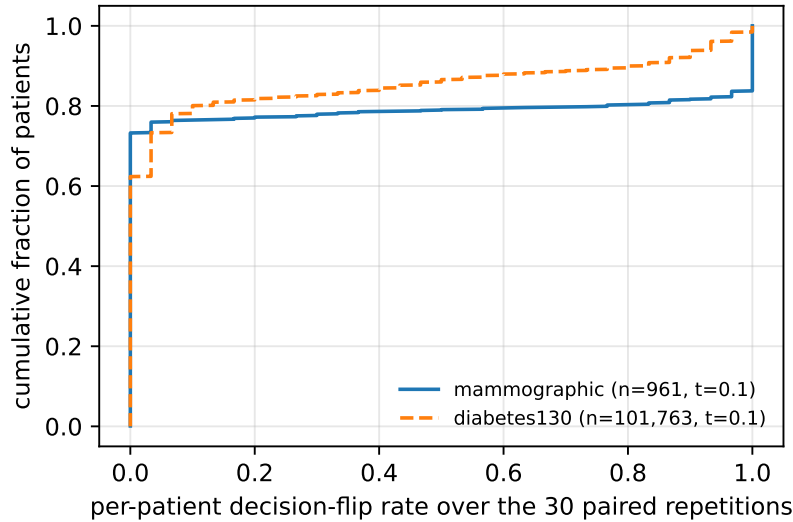


Figure 8. Cumulative distribution of per-patient decision-flip rates under aggressive reduction, at the clinical thresholds of the two worked examples. A flip rate of 1 means every one of the 30 paired models changes its decision for that patient when features are reduced.

assert that its safety cannot be presumed from discrimination or from dataset properties, and must be measured.

Practical guidance. When feature reduction is being considered, calibration (Brier and ECE), selective reliability (the risk–coverage curve), clinical net benefit (a decision curve), distribution-free set size (conformal efficiency), and subgroup gaps should be reported for each candidate budget alongside discrimination, on the same paired cross-validation splits we use here. Because the harm is heterogeneous and no dataset property we measured certifies a budget in advance, the decision must be made per dataset and per learner by *running* the audit, not by consulting a proxy for it. The retained count locates where the risk concentrates—small absolute k , most severely below $k \approx 4$ —but that identifies a high-risk regime rather than certifying a safe one. Where the calibration-error component is the primary concern, post-hoc recalibration of the reduced model is an effective and computationally inexpensive mitigation (Section 4.9), and is particularly important for tree ensembles, whose ECE penalty is largely attributable to baseline miscalibration. However, recalibration is not a substitute for retaining the features: it does not restore the selective-reliability (AURC) or Brier loss, so where the model’s confidence will be used to triage or defer, feature reduction itself must be reconsidered rather than corrected post hoc.

Certify before you cut — a six-step checklist.

1. Fix the outcomes, the candidate budgets, and the materiality rule *before* any result is inspected (pre-register them).
2. Evaluate every budget under an in-fold leakage firewall, paired against the full-feature reference on shared cross-validation splits.
3. Report calibration (ECE, Brier), selective reliability (AURC), clinical net benefit, conformal set size, and subgroup gaps alongside discrimination, per budget.
4. Express budgets as the *absolute* number of retained features k , not only as a fraction of p , and identify the smallest k at which no axis is materially worse.
5. Check the decision layer: at the operating threshold, count the patients whose treatment decision the reduction flips.
6. If only calibration error degrades, recalibrate post hoc; if selective reliability or net benefit degrades, retain the features — recalibration will not recover them.

Relation to established practice. The individual ingredients—feature rankers, calibration metrics, selective-prediction machinery, subgroup gaps—are all standard, and we deliberately use established forms of each so the protocol inherits their validity. The methodological contribution is their composition into a single pre-registered, leakage-safe certification protocol

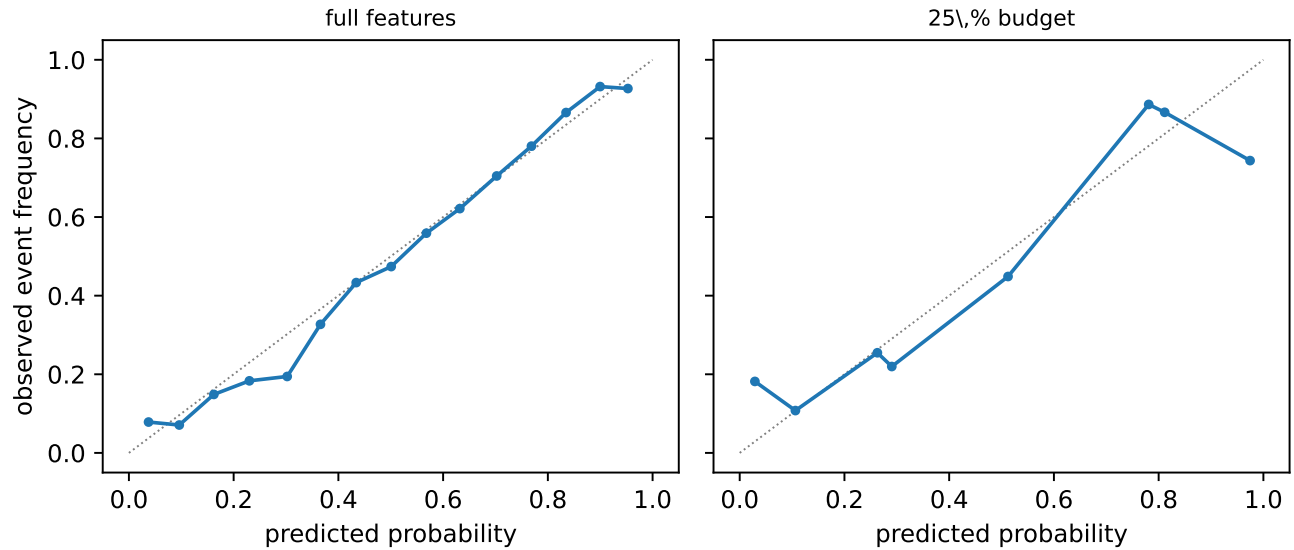


Figure 9. Reliability diagrams for the worked-example cell (mammographic mass), pooled over the 30 repetitions: full model (left) versus the one-feature reduced model (right). Reduction compresses the predicted probabilities toward the prevalence, multiplying threshold crossings near clinically relevant operating points.

that treats calibration, selective reliability, and subgroup equity as *dependent variables of the reduction level*, rather than evaluating a final model once on discrimination. It constitutes a reusable evaluation methodology for a recurring clinical-informatics decision—how aggressively to reduce features—assessed on the axes that govern deployment. We do not claim superiority for any classifier; the deliverable is the evaluation methodology and the generalizable lesson it yields.

5.1 Limitations and threats to validity

Construct. ECE is binning-dependent; we fix 15 equal-width bins a priori and treat the binning-free Brier score as the primary calibration outcome (Brier and ECE agree in direction throughout). Because that choice is a free parameter, we ran a bin-count sensitivity sweep as a separate control: a dedicated grid over the twelve datasets, all three rankers, *all three learners* and all three recalibration depths, at 5, 10, 15, 20, 30 and 50 bins under both equal-width and equal-frequency binning. The reduction penalty for the random forest is positive at every bin count under both binning schemes, and for uncalibrated gradient boosting it is materially *negative*—reduction improves that learner’s calibration—at every bin count under equal-width binning, so both of those conclusions are bin-count-robust in the direction the main analysis reports.

Exactly two cells across the sweep change sign, and neither is material. The first is the uncalibrated logistic arm under equal-width binning, which runs from -0.0021 at 5 bins to $+0.0021$ at 15; the second is gradient boosting under isotonic recalibration, which crosses between 15 and 20 bins under equal-width binning and between 20 and 30 under equal-frequency. **Neither reaches Holm-corrected materiality at any bin count**, their magnitudes never exceed 0.005, and in each case the other binning scheme places the crossover elsewhere—which is what a binning artefact looks like, and what a real effect does not. We therefore report the ECE conclusions as bin-count-robust and disclose both sign changes, rather than resolve them by choosing a binning. AURC uses confidence $|p - 0.5|$, which is a monotone function of the predicted probability itself. Selective reliability and calibration are therefore not fully independent, and are expected to move together. We accordingly do not treat AURC as separate evidence of the calibration effect: its value is that it degrades while AUROC stays flat, not that it is statistically independent of ECE or Brier. Recomputing AURC under the margin $|2p - 1|$ and under predictive entropy leaves the penalty essentially unchanged. That is a robustness check on the *parameterisation*, however, and not a test of independence, because for a binary outcome all three quantities are monotone functions of $|p - 0.5|$ and induce the same ranking by construction. We also tested the obvious model-based alternative and report that it fails for a structural reason. Taking the random forest’s *per-tree disagreement*—the spread of the base trees’ predictions on each held-out point, before any calibration wrapper—as a confidence signal gives a pooled Spearman correlation with $|p - 0.5|$ of -0.991 over 11,180 out-of-fold points, and at least 0.96 in magnitude on every one of the twelve datasets. This is mechanical rather than incidental: a forest disagrees most where its mean prediction is near 0.5 and least where it is near 0 or 1, so tree spread is a monotone re-expression of distance from the decision boundary and induces the same risk–coverage ordering. Ensemble disagreement

Algorithm 3 SUPPORTDISTANCECONFIDENCE: a confidence signal the predicted probability never enters, used to separate the part of the selective-reliability penalty that is a property of the probability scale from the part that is not.

Require: dataset (X, y) ; folds K ; neighbours m ; the cached out-of-fold probabilities p from Algorithm 1 at each budget

Ensure: a confidence $c \in \mathbb{R}^n$, and the reduction penalty measured under it

```

1: for each fold (tr, te) do
2:    $\mu, \sigma \leftarrow \text{MEAN}(X_{\text{tr}}), \text{STD}(X_{\text{tr}})$  ▷ training rows only
3:    $Z_{\text{tr}}, Z_{\text{te}} \leftarrow (X_{\text{tr}} - \mu)/\sigma, (X_{\text{te}} - \mu)/\sigma$ 
4:    $T \leftarrow \text{BALLTREE}(Z_{\text{tr}})$  ▷ held-out rows are not in the tree
5:   for each held-out row  $i \in \text{te}$  do
6:      $d_i \leftarrow \frac{1}{m} \sum_{j \in \text{NEARESTNEIGHBOURS}(T, z_i, m)} \|z_i - z_j\|_2$ 
7:      $c_i \leftarrow -d_i$  ▷ nearer the training support  $\Rightarrow$  more confident
8:   end for
9: end for
10: return  $c$ , and  $\text{AURC}(y, p, c)$  at each budget

```

Why this separates the two components. c is computed from X alone, so it shares no functional form with $|p - 0.5|$; and the full-feature and reduced models are compared under the *same* c , so any surviving penalty cannot be an artefact of the confidence being a transformation of the quantity under test.

therefore cannot serve as the independent signal, and we did not pursue it further.

Establishing selective reliability as independent evidence requires a confidence estimated *outside* the predicted probability altogether, and we ran that experiment rather than leave it as a caveat. For every out-of-fold point we computed the mean Euclidean distance d to its 10 nearest neighbours in the full feature space, standardised on training-fold statistics only, and used $-d$ as the confidence—a quantity into which the model’s predicted probability never enters. It behaves as the independent signal that ensemble disagreement was not: pooled over 106,943 out-of-fold points its Spearman correlation with $|p - 0.5|$ is $+0.40$, against -0.991 for tree spread. That pooled figure is dominated by the readmission cohort, which supplies 95 % of the points, so we report the unweighted mean over the twelve datasets, $+0.27$, as the central value. The per-dataset correlations span -0.65 to $+0.74$ with mixed signs, and a dedicated check found no trend with cohort size—Spearman between n and the correlation is -0.08 ($p = 0.80$)—so the partial independence is a property of the signal rather than of any one dataset.

Recomputing the reduction penalty with $-d$ in place of $|p - 0.5|$, on the same cached out-of-fold predictions and with no refitting, then gives the sharpest statement we can make about what selective reliability was measuring. The mean AURC reduction penalty falls from $+0.0334$ under the probability-scale confidence to $+0.0116$ under the probability-independent one. Roughly two-thirds of the measured selective-reliability harm was therefore a property of the probability scale. The remaining 35 %, however, *survives*, and survives decisively—a Wilcoxon test over the 2,160 cells gives $p = 7 \times 10^{-103}$ with a positive median—and shows no dependence on cohort size either. The concession above is accordingly upgraded from a qualitative disclaimer to a measurement: a material fraction of the selective-reliability degradation persists under a confidence that cannot see the predicted probability at all, so it is partly, though not wholly, independent evidence. Consistent with the mechanism proposed in Section 4.9, the surviving component is what one would expect if reduction damages the ordering of cases by genuine difficulty rather than merely the scale on which that difficulty is reported. We did not search for a further confidence signal that would restore more of the penalty; $-d$ was pre-specified, and its result is reported as it came.

Having established that a component of the harm is independent of the probability scale, the natural next question is whether anything mitigates it, and we answered it as an audit question rather than by proposing a method. The combination rule was fixed in the deposited pre-registration before any number was computed: rank-normalise each confidence to $[0, 1]$ within a held-out fold and take their mean. No alternative was tried and no parameter was tuned. Three arms were then evaluated on the identical cached predictions—the probability margin alone, the distance confidence alone, and their combination—and all three are reported, never the combination by itself.

The reduction penalty under the combined confidence is $+0.0173$ (95 % CI $+0.0163$ to $+0.0183$) against $+0.0335$ ($+0.0318$ to $+0.0351$) under the probability margin: intervals disjoint, and 48 % of the measured harm removed. The mitigation is positive on all twelve datasets, and largest where the baseline penalty is largest. The mitigation is nonetheless confined to one axis, and the other one moves against it. The same run measured selective net benefit at esult80 % coverage, and the combined confidence does not recover it: the reduction penalty there is esult -0.0163 (esult -0.0176 to

esult -0.0150) against esult -0.0097 (esult -0.0110 to esult -0.0085) under the probability margin alone—slightly *worse*, with disjoint intervals. A second signal therefore helps the model order its errors and does not help the net benefit of acting on that order. Since this paper treats the two as equally deployment-relevant, the distinction matters: nothing here licenses reading the esult48 % figure as a recovery of clinical utility. This is an audit finding with a practical implication, not a proposed method,

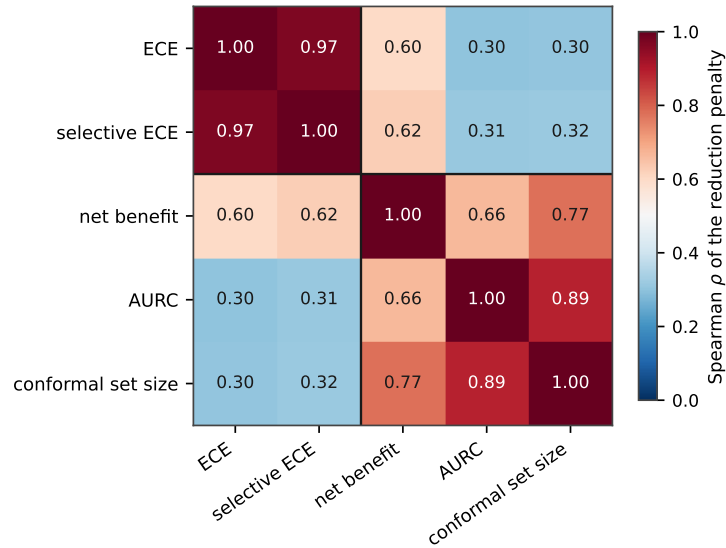


Figure 10. The reliability axes are two groups, not one scale. Spearman correlation of the aggressive-budget reduction penalty across all 108 dataset–ranker–learner cells, with every axis oriented so that positive means worse. Two blocks separate cleanly: quantities defined on the probability *scale* (calibration error, selective calibration error) cohere at +0.97, quantities defined on the probability *ordering* (selective reliability, conformal set size) cohere at +0.89, and the two blocks are nearly independent of each other (+0.30). Net benefit sits between them, as a decision rule that reads a calibrated probability at a threshold should. A single calibration number therefore cannot stand in for the audit.

and it says nothing about how such a rule would compare against a properly specified baseline under an explicit cost model. That comparison is a separate study. Two details bound the claim: the combined confidence does *not* restore the penalty to zero, so a majority of the harm remains; and the arm’s own baseline reproduces the distance–confidence measurement above to four decimals (+0.0334 against +0.0116), which is a useful check, since the two experiments were run separately and share only their cached predictions. We also asked whether that surviving component is the same phenomenon as the distributed-signal account of the high-dimensional arm—that is, whether the penalty that outlives a probability-independent confidence sits preferentially in the datasets whose signal is concentrated, or whose retained counts are smallest. It does not: across the twelve datasets the surviving penalty correlates with the concentration index at $\rho = +0.21$ ($p = 0.51$), with the retained count at -0.16 , and with p at -0.13 —all null. On this evidence the two are separate findings rather than one mechanism seen twice, though at twelve datasets the test is low-powered. Subgroup gaps use a coarse protected attribute (sex, gender, or a binary age split) and a minimum subgroup size, so they are conservative; following SAGER guidance we note that these attributes are recorded only as binary categories, that sex and gender are not distinguished in the source data, and that a finer sex- and gender-based analysis was not possible with these public datasets. The base learners are reported uncalibrated in the main analysis, which confounds the random forest’s larger ECE penalty with its known baseline miscalibration; the leakage-safe recalibration control (Section 4.9) disambiguates this and shows the forest’s extra ECE penalty is largely a recalibration artifact, while the AURC penalty is not.

Internal. The leakage firewall (in-fold imputation, ranking, scaling, fitting) is enforced by a test that permuting the labels collapses AUROC to chance. Paired, split-shared comparisons remove between-split variance from the reduction effect.

External. The twelve public datasets cover several clinical contexts but most are modest in size (only the readmission cohort is large) and all are low-to-moderate dimensional ($p = 3\text{--}44$), and none are prospective; absolute effect sizes may differ on larger or temporally split cohorts. We state the scope limit explicitly: **the guidance in this paper does not extend to the high-dimensional regime** ($p \gg 44$; genomic, transcriptomic, radiomic, or wide EHR feature spaces), where feature selection is often a statistical necessity rather than a convenience and redundancy structures are qualitatively different. That limit is now supported by measurement rather than asserted: the exploratory arm of Section 4.8 finds that in the $p \gg n$ regime reduction can be neutral, beneficial or severely harmful, that neither p nor n/p predicts which, and that the ordering follows the concentration of the ranker’s scores. We therefore decline to extend the guidance not because the regime is untested but because what governs it there is a property we have measured on four datasets and cannot yet claim to generalise. Every claim in the main analysis, including the screening criterion of Section 4.7, remains scoped to low/moderate-dimensional tabular clinical data of the kind

represented here. The protected attributes available limit the fairness analysis to the recorded groups.

Statistical. Wilcoxon plus Holm controls family-wise error within each (dataset, method, learner, outcome) family across the feature budgets; the aggregate cell counts we report are descriptive summaries across those families, not a single study-wide family-wise test. The cross-dataset meta-analysis ($n = 12$) is exploratory: the feature-count and retained-count correlations are leave-one-out robust and carry bootstrap intervals excluding zero, but with twelve datasets we treat the layer as hypothesis-generating, and the decisive evidence for the retained-count reading is the within-dataset exact- k arm, not the cross-dataset correlation. Bootstrap CIs quantify effect-size uncertainty, and we report materiality (significance *and* a CI excluding zero), not significance alone.

6 Conclusion

In a pre-registered, leakage-controlled audit on twelve public clinical datasets—including the full 101,763-encounter readmission cohort—aggressive feature reduction degraded selective reliability, clinical net benefit, and conformal informativeness at least as much as discrimination—dose-dependently, most severely for random forests, and in a set of settings while discrimination remained materially preserved—while probability calibration was mixed across datasets and the subgroup gaps moved in no systematic direction. The harm is governed by the *absolute number of retained features*: forced to a single feature, every dataset was materially harmed, including the wide ones whose fractional budgets never approach that regime, and datasets whose selections are nearly deterministic are harmed anyway. No dataset property we measured certifies a budget in advance—a strict certificate passed for one dataset at one budget—so the practical conclusion is that calibration, selective reliability, net benefit, and subgroup gaps must be audited explicitly, per dataset, per learner, and per retained count, before a reduced feature set is adopted; at the bedside, the audit’s materiality corresponds to real decision changes, flipping one in five biopsy decisions on the mammographic data and one in seven enrollment decisions across the full readmission cohort at clinically conventional thresholds. Post-hoc recalibration repairs the calibration-error component but not the selective-reliability or net-benefit loss, and is therefore a limited mitigation. Future work should include prospective and temporally split validation. All code, the hashed calibration configurations, the pre-registration, and every per-configuration summary and per-patient cache are released so that every number regenerates from a recorded seed.

Ethics statement

This study used only publicly available, fully de-identified secondary datasets and involved no intervention or interaction with human participants and no access to identifiable private information. It therefore does not constitute human-subjects research and was exempt from institutional review board approval.

Data availability

All twelve datasets are public and were retrieved in June 2026. Ten are from the UCI Machine Learning Repository, cited by dataset identifier: Cleveland heart disease (45), Statlog heart (145), SPECTF heart (96), Hepatitis (46), Indian Liver Patient (225), Mammographic Mass (161), Haberman survival (43), Breast Cancer Wisconsin original (15), and the Diabetes 130-US hospitals readmission cohort (296, <https://archive.ics.uci.edu/dataset/296>); the Heart-Failure clinical records dataset is UCI 519. The Pima Indians diabetes data (originally Smith et al., 1988) and the Wisconsin Diagnostic Breast Cancer dataset (originally Street et al., 1993; bundled with the scikit-learn library) are likewise public. The Diabetes-130 cohort is analysed in full ($n = 101,763$ eligible encounters); the $n = 6,000$ deterministic subsample is retained only as the registered robustness arm (Section 4.10). Per-dataset provenance (source, URL, retrieval date, preprocessing) is documented in the project repository. Raw files are not redistributed; the released code reproduces the exact preprocessing.

Code availability

All code, the hashed calibration configurations, the pre-registration (with a dated addendum disclosing every post-registration addition), the per-configuration result summaries, the per-threshold net-benefit caches, and a download script that fetches and SHA-256-verifies every raw dataset are available in the **public** project repository at <https://github.com/helghareeb/ns-clinical-fs> (MIT license), with a documented cold-start reproduction entry point; every number in this paper regenerates from committed code, the calibration configuration, and the recorded seed. The first-version configuration and result summaries are preserved unmodified under `config/submitted-v1/` and `results/submitted-v1/`. The repository is public and available for inspection during review, not only upon acceptance.

Sex and gender

Sex (Cleveland, Statlog and Heart-Failure heart, Hepatitis), gender (ILPD), an age stratum (Pima, Haberman, Mammographic), and race (the Diabetes-130 readmission cohort, six groups) are used in this study *only* as protected attributes for the subgroup-equity analysis and are never model features; the WDBC, original breast-cancer, and SPECTF datasets carry no protected attribute. We report between-subgroup calibration, discrimination, net-benefit, and conformal-coverage gaps as a function of feature reduction. Following SAGER guidance, we note as a limitation that most attributes are recorded as coarse binary categories, that sex and gender are not distinguished in the source data, and that a finer sex- and gender-based analysis was not possible with these datasets.

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Competing interests

The author declares no competing interests.

Author contributions

H.A.E. is the sole author. H.A.E. conceived and designed the study, implemented the software, performed the experiments, analysed the data, prepared the figures, and wrote and reviewed the manuscript.

Use of large language models

During the preparation of this work the author used an AI coding assistant (Claude Code) in order to support software-engineering and manuscript-preparation tasks. After using this tool the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Table 10. Subsample-agreement check on Diabetes-130: the reduction penalty at 25 % of features under the full cohort versus the corrected $n=6,000$ robustness arm, per ranker, learner, and outcome, with confidence-interval overlap and sign/verdict agreement. Computed from the released full-cohort and robustness-arm summaries.

Ranker	Learner	Outcome	Δ (full n)	Δ ($n=6000$)	CIs overlap	sign	verdict
l1_logistic	gb	auroc	+0.0010	-0.0037	NO	DISAGREE	DISAGREE
l1_logistic	gb	auroc	-0.0047	+0.0052	NO	DISAGREE	DISAGREE
l1_logistic	gb	brier	+0.0004	-0.0017	NO	DISAGREE	DISAGREE
l1_logistic	gb	ece	-0.0005	-0.0166	NO	agree	agree
l1_logistic	gb	net_benefit	-0.0005	+0.0018	NO	DISAGREE	DISAGREE
l1_logistic	logistic	auroc	+0.0016	+0.0001	NO	agree	DISAGREE
l1_logistic	logistic	auroc	-0.0042	-0.0019	NO	agree	agree
l1_logistic	logistic	brier	+0.0001	-0.0001	NO	DISAGREE	DISAGREE
l1_logistic	logistic	ece	-0.0013	-0.0025	NO	agree	agree
l1_logistic	logistic	net_benefit	-0.0001	+0.0001	NO	DISAGREE	DISAGREE
l1_logistic	rf	auroc	-0.0060	+0.0071	NO	DISAGREE	DISAGREE
l1_logistic	rf	auroc	+0.0260	-0.0032	NO	DISAGREE	agree
l1_logistic	rf	brier	-0.0025	+0.0014	NO	DISAGREE	DISAGREE
l1_logistic	rf	ece	-0.0268	-0.0057	NO	agree	agree
l1_logistic	rf	net_benefit	+0.0035	-0.0002	NO	DISAGREE	agree
mutual_info	gb	auroc	+0.0032	+0.0035	yes	agree	agree
mutual_info	gb	auroc	-0.0113	-0.0143	yes	agree	agree
mutual_info	gb	brier	+0.0006	-0.0003	NO	DISAGREE	DISAGREE
mutual_info	gb	ece	-0.0006	-0.0095	NO	agree	agree
mutual_info	gb	net_benefit	-0.0007	+0.0003	NO	DISAGREE	DISAGREE
mutual_info	logistic	auroc	+0.0039	+0.0025	yes	agree	agree
mutual_info	logistic	auroc	-0.0071	-0.0067	yes	agree	DISAGREE
mutual_info	logistic	brier	+0.0002	+0.0001	yes	agree	DISAGREE
mutual_info	logistic	ece	-0.0007	-0.0023	NO	agree	agree
mutual_info	logistic	net_benefit	-0.0002	-0.0001	yes	agree	DISAGREE
mutual_info	rf	auroc	-0.0071	+0.0064	NO	DISAGREE	DISAGREE
mutual_info	rf	auroc	+0.0278	-0.0276	NO	DISAGREE	DISAGREE
mutual_info	rf	brier	-0.0033	+0.0077	NO	DISAGREE	DISAGREE
mutual_info	rf	ece	-0.0299	+0.0209	NO	DISAGREE	DISAGREE
mutual_info	rf	net_benefit	+0.0043	-0.0065	NO	DISAGREE	DISAGREE
rf_importance	gb	auroc	+0.0266	+0.0185	NO	agree	agree
rf_importance	gb	auroc	-0.1091	-0.0708	NO	agree	agree
rf_importance	gb	brier	+0.0041	+0.0032	NO	agree	agree
rf_importance	gb	ece	-0.0006	+0.0055	NO	DISAGREE	DISAGREE
rf_importance	gb	net_benefit	-0.0052	-0.0040	NO	agree	agree
rf_importance	logistic	auroc	+0.0200	+0.0167	NO	agree	agree
rf_importance	logistic	auroc	-0.0839	-0.0656	NO	agree	agree
rf_importance	logistic	brier	+0.0026	+0.0020	NO	agree	agree
rf_importance	logistic	ece	-0.0012	-0.0019	yes	agree	agree
rf_importance	logistic	net_benefit	-0.0034	-0.0027	NO	agree	agree
rf_importance	rf	auroc	+0.0299	+0.0171	NO	agree	agree
rf_importance	rf	auroc	-0.1092	-0.0766	NO	agree	agree
rf_importance	rf	brier	+0.0221	+0.0147	NO	agree	agree
rf_importance	rf	ece	+0.0724	+0.0491	NO	agree	agree
rf_importance	rf	net_benefit	-0.0234	-0.0170	NO	agree	agree

Table 11. Exemplar patients from the worked example (mammographic mass; mutual-information ranker, logistic regression; biopsy threshold $t = 0.10$). $\bar{p}$ is the mean out-of-fold predicted malignancy risk over the 30 paired repetitions under the full model and the one-feature reduced model; the flip rate is the fraction of repetitions in which the two models disagree at t . Patients with malignant masses pushed below the biopsy threshold are listed first. Computed from the released per-patient prediction caches.

Pt.	birads	age	shape	margin	density	y	$\bar{p}_{\text{full}}$	$\bar{p}_{k=1}$	treat@0.1 (full)	treat@0.1 (reduced)	flip rate
406	3	65	4	5	3	1	0.473	0.040	yes	no	1.00
614	3	46	–	5	–	1	0.176	0.057	yes	no	0.97
883	3	48	4	4	3	1	0.225	0.066	yes	no	0.97
662	3	50	–	–	3	1	0.108	0.039	yes	no	0.63
954	4	35	2	1	3	0	0.061	0.291	no	yes	1.00
929	4	22	1	1	3	0	0.020	0.291	no	yes	1.00
927	4	20	1	1	3	0	0.019	0.284	no	yes	1.00
922	4	27	2	1	3	0	0.043	0.292	no	yes	1.00

Table 12. Decision-flip rates by race group on the full Diabetes-130 cohort at the enrollment threshold $t = 0.10$ (mutual-information ranker, logistic regression). Instability is not concentrated on minority groups in this cohort. Computed from the released per-patient prediction caches.

Race group	n	mean flip rate	% flipped in majority of reps
Caucasian	76,099	0.1493	14.43 %
Unknown	2,271	0.1441	13.87 %
Other	1,505	0.1375	12.82 %
AfricanAmerican	19,210	0.1368	13.24 %
Asian	641	0.1190	11.70 %
Hispanic	2,037	0.1122	10.51 %